Dicty Stock Center
Standard Operating Procedures
(2021 V 2)

Olivia Martin, Lina Fouad and Petra Fey

Northwestern University
303 E Superior Street
Simpson Querrey 5-500
Chicago, IL 60611

Table of Contents

Introduction	3
Lab Locations & Maintenance	4-7
Media & Antibiotics	8-9
GitHub	10-11
Plating and Growing Strains	12-21
Plating and Growing Plasmids	21-23
Shipping Orders	23-25
Payment Procedures (Billing)	26-29
Storing Strains	30-34
Storing Plasmids	35
Molecular Procedures	35-39
eLab Inventory	39-40
DSC Curation	41

Introduction

The Dicty Stock Center (DSC) is part of the *Dictyostelium* community resource (DCR) that contains dictyBase housing *Dictyostelium* genomic and functional data, strains and phenotypes. The DSC serves as a central repository for *Dictyostelium discoideum* strains related species, plasmids, and bacterial food sources. The DSC fills orders from labs around the world. Additionally, we are responsible for maintaining the current inventory and overseeing its expansion with the addition of new materials submitted from users. DSC information is available online.

Dictyostelium discoideum is a species of soil amoeba that feeds upon bacteria such as *E. coli* and *K. pneumoniae* in the laboratory. It is classified as a model organism because it contains many genes that are homologous to those found in higher organisms, making it useful for genetic, cellular, and developmental research.

While *Dictyostelium discoideum* undergoes various life cycles (ex. vegetative, social, and sexual cycles), its development into a multicellular organism, through the social cycle displayed in Figure 1, is most pertinent to procedures carried out in the DSC. Upon adverse conditions such as starvation (e.g. consuming all the bacteria on a plate), single cells aggregate through cAMP signaling and eventually form a mobile slug that moves via phototaxis and thermotaxis, and ultimately form a mature fruiting body. The fruiting body consists of a stalk and a head of durable spores. About 20% of cells will become vacuolized and produce cellulose to make a stiff stalk, dying in the process. The remaining 80% of cells will form viable spores, which are durable and hatch into amoeba once conditions are favorable again. Because some cells are sacrificed in the process of species survival, *Dictyostelium* is considered an altruistic organism.

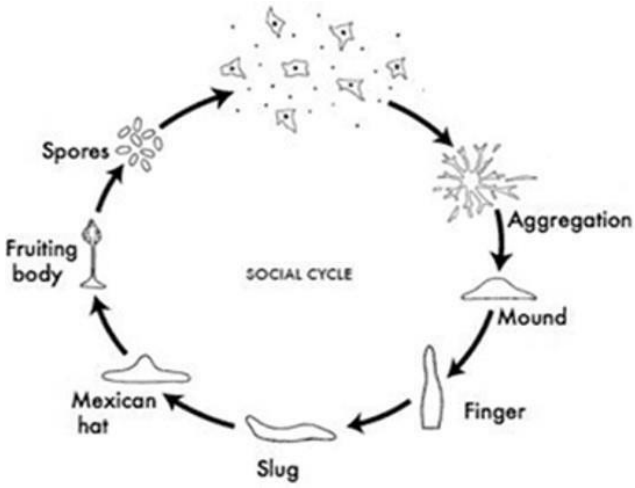


Figure 1. Developmental steps in the *Dictyostelium discoideum* social cycle

All 'wild type' cells (e.g. NC4, DdB, DH1, AX1, AX2, AX3, AX4, etc.) are able to form spores, however, we handle many mutants in the DSC. While some mutants have not lost the ability to sporulate, others have had essential genes and pathways interrupted and may therefore be arrested at any stage during development. Spores are the most durable form, thus sporulating strains are typically easier to grow and store than non-sporulating strains.

All wild type strains in the DSC are able to grow on bacterial plates. However, many years ago scientists sought to grow *Dictyostelium* without bacteria to eliminate bacterial DNA, RNA and proteases in preparations. They selected for strains that could grow in a rich nutritious broth. The resulting strains were called 'axenic' and today we have AX2, AX3, AX4 and their descendants. These strains grow in HL5 as well as on bacteria. HL5 is an axenic, rich liquid media, in which axenic cells grow in the absence of bacteria feeding by macropinocytosis. Many, but not all, mutants in the DSC are descendants of th ese axenic strains, and therefore may grow in HL5. Strains grow in HL5 as single layer cells in a petri dish and as long as they feed and divide there is no development (fruiting body formation). Axenic cells also grow in a shaking flask, often called "in suspension." Growth in suspension is used in the DSC only for analysis as it is the least natural habitat. *Dictyostelium* cells will be more fragile growing axenically, especially in suspension.

Lab Locations & Maintenance

Material Locations

Strains: Strains are typically located in the liquid nitrogen tanks in three replicates. Tank A and Tank B are located in the common equipment room in Simpson Querrey (SQ) 5-419. Tank C is located in the basement of the Olsen/McGaw Pavilion in the McGaw Freezer Farm (need special access on key card – contact Kiara Moore-Meza for help).

Plasmids: Plasmids stored as DNA are located in Styrofoam boxes in the -20°C freezer located in SQ 5-
507. All plasmids stored as bacterial transformants are kept in the -80°C freezers located in the common equipment room SQ 5-419. The primary set of plasmids is located in Freezer A, shelf 3. Duplicates of these plasmids are located in Freezer B, shelf 3. The goal is to have all plasmids as DNA and bacterial strain available.

Antibodies: There is one set of antibodies, located in -80°C Freezer B shelf 3.

Genomic DNA library: These libraries are stored in 96-well plates located in labeled Ziploc bags in Freezer B.

Original GWDI strains: GWDI strains in 96 well plates are located in -80°C Freezer B.

Dicty House

All *Dictyostelium* strains grow best in slightly humid environments. To create such an environment for strains growing on bacterial plates, cover the entire bottom of the clear Dicty container with three layers of moist paper towels. Use the water bottle to keep the paper towels consistently moist. Every two weeks clean the box with soap and water and replace the paper towels.

NOTE: *The container is clear because Dicty development needs light. In nature, the motility of the slug guarantees that fruiting body formation occurs near surface areas, where wide spore dispersal is facilitated.*

Nitrogen Tank Refills

Liquid nitrogen tanks must be filled up every two weeks or sooner if running low. To order nitrogen refills, go to nugastanks.com, and under Chicago Campus select “Order a Gas Tank.”

For Tank A and B order 45 liters. For Tank C order 15 liters, unless it seems depleted, then choose 30 liters. For Tank A and B location enter SQ 5-419 (the common equipment room or LER), for Tank C enter Olson 0-250.

NOTE: *Make sure to update the contact information for the following personal is located on each freezer and liquid nitrogen tank: full-time research technologist, part-time student worker, and the lab manager.*

Freezers

Freezers need to be maintained by defrosting freezers when ice buildup becomes extensive or more than 3 cm thick. Additionally, air condensers (sponges located behind the grate on the bottom front of the freezers) are cleaned at least once a year to remove dust. Clean air condenser in sink and allow enough time to dry before returning. Freezer maintenance contract is maintained with Environmental & Temperature Solutions (ETS) https://www.envirotempsolutions.com/.

Dry Ice

Dry ice available to CGM on the 5th floor and is located in a labeled gray container in SQ 5 -500. If this stock of dry ice runs out, more dry ice can be obtained from the SQ loading dock, accessible via the freight elevators. For this, the current grand chart string number is needed. Contact Sandra Look at sandra-look@northwestern.edu or Taurean Egan-Henderson at t-egan@northwestern.edu.

Media

Media ingredients and equipment are located in SQ 5-507. Our materials are located on the shelves to the left. When new ingredients/chemicals are received, mark the bottle with the date of receipt and 'DSC'. Prepared agar plates are kept in the refrigerator.

Dishwasher

The main dishwasher is in SQ 5-500 and there is another large dishwasher in SQ 5-419. The autoclave and dry sterilizer are in the common equipment room SQ 5-419.

Autoclave

Cycle	Temperature	Sterilization Time	Dry Time
Liquid	121 C	25	NA
		45	
		35	
Vacuum	121 C	15	15
		45	20
		60	30

Table 1. Autoclave cycles

Autoclave Cycles.

For covered glassware (ex. empty beakers, flasks covered with foil), use the Vacuum cycle. For liquids (e.g. media), fill the autoclave bin with a shallow layer of water and use the Liquid cycle set for 45 minutes. However, for HL5, use Liquid set at 25 minutes. (see Media & Antibiotics-Maltose HL5, page 7). Make sure jacket is on and at about16 PSI – if not, press touch screen to turn jacket on and wait 20 mins. On the touch screen, press 'Cycle Select' password: 111, press 'Select Type' (shown below), make sure door is close 'Press Door Close' and hold and press 'Run Cycle'.

NOTE: *Autoclave sterilization time settings can be adjusted before each cycle is started.*

Dry Sterilizer

NOTE: *Make sure the power and heat switches are on – if not, turn the power switch on first and wait for 30 seconds to load, then turn on the heat switch.*

On touch screen, press 'Profile Actions', press 'Profiles' and select profile, press 'Actions' and press 'Run Profile'. Run time is about 2 hours. Door can be pulled open at any time. Use this cycle exclusively for sterilizing dry items (ex. empty beakers, flasks covered with foil).

Profile	Temperature
Sterilizing	121 C
Sterilizing	160 C
Dehydrogenation	Unknown

Table 2. Dry Sterilizer cycles

NOTE: *The dry sterilizer uses different sterilizing tape than the autoclave tape. The autoclave indicator tape will not work in the dry sterilizer.*

General Lab Equipment

A microcentrifuge and the medium centrifuge are located on the benchtop. The Avanti (cold) centrifuge, located in SQ 5-419, is also available for larger, temperature-sensitive items for everyone on the 5th floor.

A 37°C incubator for plates is located on the benchtop.

A cell density meter that takes optical density measurements is stored in the box above the benchtop. A small culture shaker e.g. for suspension growth, is located on the benchtop.

We do not own a speed vacuum, but one is available by the small dishwasher near SQ 5-507.Lab Safety

All lab safety materials can be accessed at https://lumen.northwestern.edu/. Furthermore, the full- time technician needs to be added as the lab safety designate by the PI. After being added as a designate, the technician can manage the lab's safety profile and assign training for new lab members.

NOTE: Every fall a laboratory safety review is done and is scheduled by the Office of Research Safety.

Hazardous Waste Disposal/ Glassware Disposal

Most biohazardous waste generated in the lab can be discarded in double-bagged red biohazard bags (25" x 35" for bins). Appropriate biohazard waste for biohazard bags include cell culture plates, discarded agar, cryovials etc. When bags are full, they are deposited in the large red bin in SQ 5 -419. Label discarded bags with tape detailing "DSC – Chisholm, [Date], [Cell Culture Plates/Serological Pipette Box], SQ-500".

For liquid biohazardous waste, add bleach to the container and let it sit for 30 minutes before draining in the sink. Flush with plenty of water to dilute the bleach.

Serological pipette tips are discarded in burn boxes obtained from the stock room located in the Searle Medical Research Building, Room B-652. Biological and chemically contaminated sharp objects (including small pipet tips) are collected in wide mouth jars obtained from the stock room. Glass disposal bins can also be picked up from the stock room. When boxes/containers/bins are full, they are marked as trash to be picked up. Any further questions can be directed to researchsafety@northwestern.edu.

Hazardous Waste Removal

https://researchsafety.northwestern.edu/hazardous-waste/hazardous-waste-disposal-guide-purple-guide

1. Access Lumen
2. Select 'Waste'
3. Click 'Submit New Chemical Waste Pickup Request'

NOTE: *UV light bulbs can be picked up as hazardous waste.*

UV Trans-illuminator

DNA stained gels are visualized in the McNally gel room SQ 5-510. Make sure to put on full PPE before turning on the UV lamp. Wear a lab coat, gloves and a face shield. Since we are using a nontoxic DNA stain, use a smart phone or digital camera to take a picture of the illuminated gel for documentation.

Ordering Lab Supplies

All lab materials can be ordered through Fischer-Sci at https://www.fishersci.com/us/en/home.html. To place an order, you will need to search for your materials, then compose an email in a list format including the following information for each item:

1. Quantity
2. Full Product Name
3. Catalog Number
4. URL

Send order emails to Michelle Mohney at michelle.mohney@northwestern.edu , who will place the order.

Other items not ordered through Fischer-Sci can also be sent to Michelle in a similar manner. For example, cardboard boxes for shipping can be ordered here: https://www.packagingsupplies.com/brown_kraft_1-piece_folding_cartons.html (product ID C342H6.)

Media & Antibiotics

All media and buffers are autoclaved using Liquid Cycle with a 35-minute sterilization time. Maltose HL5 needs Liquid Cycle with a 25-minute sterilization time because sterilization over 25 minutes destroys vitamins in the medium (see Lab Locations & Maintenance,Table. 1, p. 5). Sterilization times may be adjusted before each cycle runs. Lab materials are listed at dictyBase: (https://dictycr.org/research/techniques/media).

SOC (1 Liter):

20 g Tryptoneg Yeast Extract 2 mL 5M NaCl
2.5 mL 1 M KCl

LPB Agar (Lactose, Peptone, 1 liter):

1 g Lactose
1 g Bactopeptone
5.1 g Na_2HPO_4*7 H_2O
4.2 g KH_2PO_4 20 g Agar

GYP Agar (Glucose, Yeast, Peptone, 1 liter):

1 g Glucose
2 g Bactopeptone
0.25 g Yeast Extract
5.1 g Na_2HPO4*7 H_2O
4.2 g KH_2PO_4 20 g Agar

Maltose HL5 (1 liter):

14.3 g Bactopeptone
7.15 g Yeast Extract
18 g Maltose Monohydrate 0.641 g Na_2HPO_4*2H_2O
0.490 g KH_2PO_4
0.05 g dihydrostreptomycin-sulfate

L Broth / Agar (1 Liter):

10 g NaCl
10 g Tryptone
5 g Yeast Extract
*For LB Agar add 20 g agar

10X KK2 Buffer (1 Liter):

22 g KH_2PO_4 (monobasic) 7 g K_2HPO_4 (dibasic)

30% Sucrose in 10 mM KK2 Buffer (50 mL):

12 mL 10X KK2 Buffer 25 mL 60% sucrose
3 mL sterile dH_2O

Maintain phenotypes

To keep phenotypes tight for species such as D. minunutum you may add 1.25 g charcoal powder to 250 mL LPB.

Add Strep or combine with Pen

Streptomycin or in combination with Penicillin (Pen/Strep) is added to reduce bacterial contamination. Streptomycin can be added as dihydrostreptomycin-sulfate powder before autoclaving or as sterile streptomycin solution after autoclaving to a final concentration of 300 µg /ml. Penicillin is added to a final concentration of 100 µg /ml (Antibiotics- Table 3, page 8).

SOC pH

Adjust pH to 7.0 with KOH
After autoclaving add 1/100 volume of 2M Mg2+ and 1/100 volume of 2M glucose

30% Sucrose

Prepare a 60% stock sucrose solution by adding 60 g sucrose/100 mL dH_2O, then autoclave
10X KK2 Buffer, 60% sucrose solution, and water should all be combined in the hood to avoid contamination

NOTE: Media can be stored at 4 °C for several weeks. When medium contains drugs, use within weeks as the drugs may degrade. Therefore we add drugs to medium in 50 ml tubes when not much is needed.

Antibiotic Concentrations

Antibiotics are stored at -20°C in aliquots of 1 mL to avoid repeated freeze-thaw cycles. The table below shows the concentrations bacterial (LB plates) and *Dictyostelium* strains are grown at. To determine the amount of antibiotic stock solution to add to the medium, use the equation $C_1V_1=C_2V_2$ (C_1 is the stock solution concentration, V_1 is the volume of stock solution to add to media, C_2 is the final antibiotic solution concentration, V_2 is the volume of the medium). Antibiotics are added after the medium has been autoclaved and cooled for at least 20 minutes in a 65°C water bath.

Antibiotic	Medium	Stock Concentration in H_2O	Final Concentration	Volume to ~10mL	Volume to ~25mL	Volume to ~250mL	Volume to ~1 L
Ampicillin	LB	100 mg/mL	50 µg/mL	--	12.5 µL	125 µL	500 µL
Kanamycin	LB	20 mg/mL	50 µg/mL	--	62.5 µL	--	--
Pen/Strep	HL5	10,000 units (Pen) + 10 mg (Strep) / mL	100 units (Pen) +100 µg/mL S.	100 µL	250 µL	2.5 mL	--
Penicillin	HL5	100 mg/mL	100 µg/mL	1 µL	--	--	--
Streptomycin	HL5	300 mg/mL	300 µg/mL	0.67 µL	--	16.7 µL	--
Blasticidin	LPB	10 mg/mL	20 µg/mL	--	50 µL	--	--
	HL5		10 µg/mL	10 µL	--	250 µL	--
Neomycin (G418)	LB	50 mg/mL	60 µg/mL	--	--	300 µL	--
	LPB/GYP		30 µg/mL	--	15 µL	--	600 µL
	HL5		20 µg/mL	4 µL	--	100 µL	--
Hygromycin	LPB	50 mg/mL	60 µg/mL	--	30 µL	300 µL	--
	HL5		50 µg/mL	10 µL	25 µL	--	--
Chloramphenicol in EtOH	LB	25 mg/mL	25 µg/mL	--	25 µL	250 µL	--
Kanamycin	LB	100 mg/mL	50 µg/mL	--	12.5 µL	125 µL	--

Table 3. Antibiotic solutions and concentrations

NOTE: *Prepare drug solutions in sterilized dH2O and filter sterilize before storing.*

Creating Antibiotic Stock Concentration Aliquots

1. Determine appropriate amount of antibiotic needed for stock solution (refer to antibiotic table above)
    a. ie. For Kanamycin stock solution of 100 mg/mL, for 4 mL we will need 400 mg and 4 mL, so label 8 microcentrifuge tubes
2. Determine the amount of sterilized distilled water or ethanol that is needed to create the stock concentration
3. Appropriately label stock concentration aliquots with name of antibiotic, concertation, and date
4. Add antibiotic to 50 mL autoclaved beaker
5. Add water or ethanol to beaker with serological pipette
6. Mix stock solution by pipetting solution up and down
7. With a sterile syringe, take up antibiotic stock solution
8. Screw on a 0.22 µm filter onto a sterile syringe and then, dispense 500µL of antibiotic into microcentrifuge tubes

NOTE: *DO NOT touch bottom of syringe and keep the filter in the plastic covering as you screw on the syringe.*

GitHub

GitHub (https://github.com/dictyBase) is used to keep track of the **order** and **payment status**. A GitHub account is necessary to use the resource.

There are two repositories in GitHub relevant to the DSC: **Stock Center Orders** and **User Requests**.

A. **Stock Center Orders** is the tracker primarily used by the DSC, as this contains each order and keeps track of the order status.
B. **User Requests are** used when users have general questions before placing an order or need help after receiving the materials. If this user request relates to an order placement, link the issues in the two repositories by copy/pasting the link of one issue into the comments section of the other. Add the Stock Center label (see below about labels).

Below is a sample of what a DSC Order issue in the Stock Center Order repository looks like

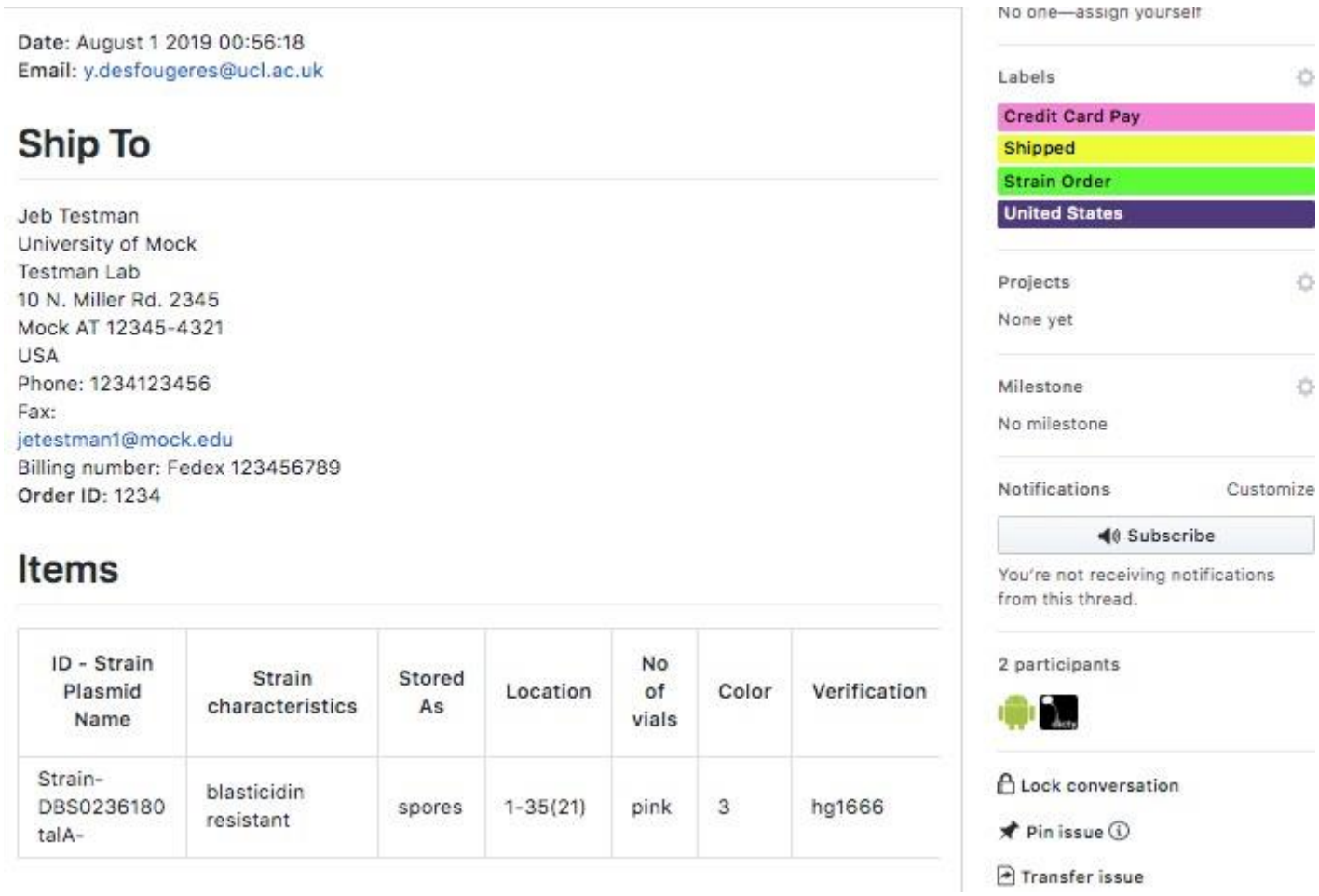


Figure 2. Sample order in Stock-Center-Orders repository

Creating New Issues

When a user places a new strain/plasmid order, a confirmation email is sent automatically to members of Dicty Stocks. These orders are automatically added into GitHub including labels such as Material (strain, plasmid), Country, and Payment Method.

NOTE: *Additional materials such as Antibodies, DNA libraries, and GWDI pooled libraries cannot be ordered through the database because they are limited. These orders need to be added manually with the shipping information and labels such as* Antibody Order, A Library Order, GWDI Pooled Library. *Because these orders do not go through the database, there will be no order ID except the one created in GitHub.*

Labels

In addition to identifying the country and material type, labels are used to track orders' growth, shipment, and payment status. Refer to the Plating Orders, Shipping Orders, and Order Payments sections below to see when to add these frequently used labels.

The following labels are less common and to be used when appropriate:

Axenic Medium - used to indicate that a user has requested their strain be shipped in HL5.

On Hold - used when a user has placed an order but wants it shipped later.

Rejected/Canceled - used when a user cancels their order. These issues can be closed immediately.

No Charge - used when a user will not be charged for their order due to an error on our account. These orders are not added to NU Core, and thus bypass the payment status steps. They are closed once they have been shipped. "No charge" orders are not to be confused with "waiver/no payment" orders.

'Manual' Order - used when we have placed the order for a user through the DSC webpage (i.e. when user emails a list of strains or orders the materials that are not in the database.)

NOTE: *Manual orders can be added under your name or the user's, depending on how much information you have when placing the order. If the order is under your name, edit the name and shipping information and forward the order confirmation to the user if it was not sent to their email.*

Alert - used when there is a problem with a user's order. This label can be applied for a variety of situations such as difficulty growing a strain or missing/conflicting shipment or payment information.

GITHUB

After plating an order, add the Growing / in Preparation label.

Comments

An order's shipment date, tracking ID, and delivery status is recorded in the comments section of each issue.

Additionally, any further communication with users is to be recorded in the comments section. For example, if you have emailed a user informing them of a delay in their order when it has not been shipped within 10 days, record your notification, as well as their response.

Plating and Growing Strains

ID – Strain / Plasmid*:* The ID is a unique identifier. Both, strains and plasmids have unique IDs. Plasmids are typically named as submitted. If no lower case 'p' precedes the name, we add it for identifying the item as plasmid. For strains, the strain descriptor provides a detailed description. Depositors may give strains different and/or shorthand names in publications.

NOTE: *If the strain descriptor is too long or common, the tube may be labeled with the systematic name or any other name instead. In the strain profile, check '***Strain Names'** *for other names used to label the stock.*

Strain characteristics*:* This field contains information about the antibiotic resistance of the mutant strain. The drug(s) should be contained in the growth medium or the kind of mutation and even if a mutated strain cannot grow axenically. Typical wild type strains do not have any characteristic as those have not been manipulated.

Stored As: This determines how to grow the strain. If stored as 'spores' or 'cells' the strain will be plated on agar on a bacterial lawn. If 'axenic' the strain will be plated in HL5, often a spot is also made on a prepared bacterial plate with the last drop in the tube.

Location: All regular strains are stored in nitrogen tanks. Typically, the strains are taken from Tank B (closest location), unless the stock in Tank B is low or no longer viable, in which case we use Tank A. Tank C is a backup tank (and in a different building), and should only be used as a last resort. If you are using the **first** tube from Tank C, make sure to restore this strain immediately, and after viability tests replace the stocks in all tanks. If there are still extra viable older stocks in Tank C after re -storage, those might be kept temporarily in the box 'Tank C - Restore' at a -80°C freezer, shelf 3 and used as soon as necessary.

In liquid nitrogen, the first number under the location indicates the crane number (cranes 1-6), the second shows the box number, and the number in parentheses is the position of the tube within the box. After locating the tube, verify that color and name match the information on the order form.

NOTE: *All Dicty strains grow at room temperature (21-23°C), no special incubators needed!*

Checking Strain Phenotypes

Make sure to look up phenotypes in dictyBase and compare with your observations prior to shipping the strains. It is important to not just look at the phenotypes of the strain ordered, but to check other mutants with the same gene mutation (e.g. compare knock-out strains with other knock-outs in the same gene).

Phenotypes

1. Search the DBS ID, the Strain descriptor or the strain summary from the strain catalog.
2. From the Strain details page, record any developmental phenotypes (i.e. decreased cell motility, abolished aggregation) on order sheet
3. If there are no phenotypes listed or no developmental phenotypes, click the Associated Gene(s)
4. On the gene page, check the 'Strains and Phenotypes' section for similar strains and record the relevant phenotypes on order sheet next to each strain

NOTE: *When inferring phenotypes from similar strains, it's necessary to read the papers to judge if phenotypes between different strains are expected to be the same or similar. The strain descriptor helps to distinguish phenotypes from other strains. Also, contact Petra if you have further questions.*

Genotypes

If similarity of strains for phenotype analysis cannot be recognized from the strain descriptor, it might help inspecting the genotype. The genotype exactly describes the genetic makeup of a strain and is annotated in the strain record, e.g. for strain DBS0351349:
axeA2,axeB2,axeC2,racE-,[bsR-loxP],racE(G23V), pTX-GFP],**bsR,neoR.**

The axeA2 string means AX2 is the parental strain, the gene mutations are listed next with constructs used in brackets, and at the end are the drug resistances (non-italic).

Strains that don't grow/ wrong phenotype
If the strain does not initially grow, you must test all of the available inventory. It is important to accurately mark each plate tested with inventory location and record results.

1. Test remaining inventory
2. If still no growth even after plating the backup, contact original depositor to resubmit strain:

Dear Dr. [**depositor name**],

We are having some difficulties with the strain, [strain name], that you submitted some time ago to the stock center.
[**strain name**] – [**strain profile link**]

After testing the strain from all available inventory, we have not seen any growth. There may have been an issue when the strains were originally stored.

If you still have stock of the strain and could resend it to the DSC, that would be great and we would pay for shipping.

Here is our shipping information: [**your name**]
Northwestern University Simpson Querrey 5-500 303 E Superior Street Chicago, IL 60611

FedEx Account #: 1230-3428-7 Tel. (312) 503-4169

Best wishes,
[**your name**]

NOTE: *Consider other related, available strains that the user may be interested in. The parent, mutagenesis method, and genotype are important factors to compare to potential other strains, or if unavailable and parent and plasmid are available the strain can be recreated by the researcher and resubmitted to the DSC.*

Bacterial Growth

Plate Types

We typically use two types of plates in the DSC: LPB and GYP agar. LBP plates are used by default for all bacterially plated orders; GYP is a richer medium and is used for strains that are difficult to grow or for storage (see Storage Procedures - Storage from Bacterial Plates, page 30-32).

Always keep LPB and neomycin LPB plates in stock. Strains with no antibiotic resistance, such as natural isolates, are plated on plain LPB plates. Other strains might be resistant to neomycin, blasticidin, or hygromycin, etc.). Neomycin resistant strains are of most concern because the resistance is easily lost. These strains **must** be plated on neomycin-containing plates.

For the purposes of short-term plating, strains with other types of resistances can be plated on plain LPB plates. However, if a strain takes considerably longer than 4 days to grow, then supply the medium with the appropriate antibiotic.

NOTE: *To add a drug solution to a solidified agar plate, estimate the volume of the agar (usually about 30 mL). Calculate the drug amount and dilute to a final volume of 300 µL. With a sterile spreader distribute evenly and let the plate sit for several hours for the drug to penetrate the agar.*

Bacterial Food Source

E. coli B/r and *E. coli* B/r neoR are required as a food source for strains growing on agar plates. The B/r strain is a UV treated isolate of *E. coli* strain B and is easier for *Dictyostelium* to digest than wild type *E. coli*. *E. coli* B/r is used for plain LPB/GYP plates; *E. coli* B/r neoR is used for neomycin LPB/GYP plates.

Prepare a plain LB plate and a neomycin LB plate for *E. coli* B/r and *E. coli* B/r neoR respectively and streak out each bacterial strain to obtain single colonies (see Plating Orders - Figure 10, page 21). Incubate at 37°C overnight, seal with parafilm and may be stored at 4°C up to 6 weeks.

To initiate a liquid culture of each bacterial food strain, inoculate a single colony from the streak on the LB plate into 30 mL LB or neomycin LB in a sterile flask. Grow the culture in a shaking incubator at 37°C overnight. These cultures are stored at 4°C for up to 2 weeks.

NOTE: *E. coli B/r and E. coli B/r neoR are used as the default food source and should always be kept on hand. Klebsiella aerogenes (K.a.) and other bacterial strains can be used as a food source upon request or as an alternative for problematic strains.*

Plating a Bacterial Lawn for Dictyostelium growth

1. Label a plain or neomycin plate with the strain ID, strain name, and date.
2. Spread 250-300 µL of the appropriate food bacteria onto the plate using a sterilized glass spreader.
3. Carefully dry the plate under the hood (do not over dry!).
4. Retrieve the strain(s) from the liquid nitrogen tank and place the tube(s) on dry ice.
5. Using an ethanol-flamed sterilized screwdriver, dig into the frozen culture and streak the strain onto the center of the plate.

NOTE: *Do not let the tube warm up because the freezing agent DMSO is toxic to the cells.*

6. Grow the strain upside down in a humid 'Dicty house' for 4-7 days. Seal the plate with parafilm before shipping.

NOTE: *Always verify the expected phenotype(s) to ensure that strains have not lost and/or accumulated a mutation. Check the dictyBase strain record for available phenotype data.*

Axenic Growth

In the DSC, we use maltose HL5 for all axenic growth. If a strain has no antibiotic resistance, it is grown in plain HL5 supplied with streptomycin (see Media & Antibiotics - Maltose HL5, page 7). All drug resistant strains must be cultured with the appropriate antibiotic after the initial recovery from frozen storage (see below) because cells growing in HL5 mutate easily (see Media & Antibiotics - Table 2, page 8).

NOTE: *Check that the listed DSC drug concentrations match that in the paper reference. If concentrations are considerably different, adjust the concentration. This is especially important for neomycin resistant strains as the concentration influences copy numbers of the transformed plasmid.*

Initiating Axenic Culture

Strain that has viable spores

1. Prepare small petri dish with 3 - 5 mL of plain HL5
2. Under dissection microscope, carefully collect a couple spore heads with 200 µL pipet tip while avoiding touching the agar (the plastic tip electrostatically attracts spores)
3. Carefully dip the pipette tip into the HL5 (only open lid partially and tilt small petri plate slightly)
4. Check after a few hours or the next day for hatched cells

Strain does not aggregate or form viable spores

1. Prepare GYP harvest plates (See Storage from Bacterial Plates/Cells page 30)
2. Harvest and wash cells **at least** 4 times to ensure all bacteria are removed (supernatant is clear).
3. Count cells and prepare the number of HL5 plates based on having 1.25×10^4 cells per plate

NOTE: *For the remaining steps, follow either method the Pen/Strep or KK2 method*

Pen/Strep

4. Each petri dish should have 10 mL of HL5 + penicillin and streptomycin (100 units pen/100 ug strep per 1 mL of HL5)
   NOTE: *Refer to the Pen/Strep section of the antibiotics section above.*
5. Within at least 4 hours of plating, change the media to the appropriate HL5 + drug or fresh HL5 + pen/strep if no antibiotic resistance

KK2

4. Add cells to petri dishs with 10 mL of KK2 Plate cells for 4-5 hours in KK2 so they may consume the remaining bacteria.
5. Between 4 – 5 hours, change KK2 buffer to Pen/Strep HL5. Check the cells every hour. (If cells start to elongate it's urgent to move them into HL5
   NOTE: *At about 6 hours starving cells start developing, produce cAMP and elongate. Cells migrate and aggregate. Do not leave them in buffer for longer than 5 hours.*

6. **Continue to check plates to monitor for bacterial growth and replace HL5 daily**

NOTE: *If too many bacteria are still present in liquid culture, they will outgrow the Dictyostelium in the rich axenic medium, then the entire culture must be discarded, and the process needs to be restarted.*

Preparing Axenic Culture from Frozen Inventory

Day 1: Transfer Cells to HL5

1. Take plain HL5 from the fridge and warm to room temperature. Label a new 10 cm petri dish with the strain name and date.
   NOTE: *All strains are cultured in plain HL5 initially to recover from frozen stage. Antibiotics, if applied, are added the next day.* Add 10 mL of HL5 to the dish.
2. Quickly thaw an entire tube of the axenic strain by swirling in a 37°C water bath. Check the tube often and remove when there is a small amount of ice left. Do not let the tube warm up!
3. Immediately transfer the entire contents of the tube to HL5 and gently swirl the dish to disperse the cells. Surviving cells will attach to the bottom of the dish.
4. Spot the last drop in the tube onto a prepared bacterial plate. This allows to ship the user an axenically and bacterially grown strain and/or store the strain as 'cell' or 'spore' stock.

NOTE: *After a few hours check the cells. If only some survivors have attached but there is a lot of debris floating, change the medium once on the same day to remove the dead cell debris and the DMSO early.*

Day 2: HL5 Media Replacement & Antibiotic Addition

1. Carefully remove the medium without disturbing the attached cells by tilting the plate slightly and pipette the medium out into a beaker in order to rid the medium of DMSO and remove dead cells, if present.
2. Gently add fresh HL5 to the cells (preferably at the same spot at side of dish) and gently swirl plate to disperse any cells that were dislodged in the process.

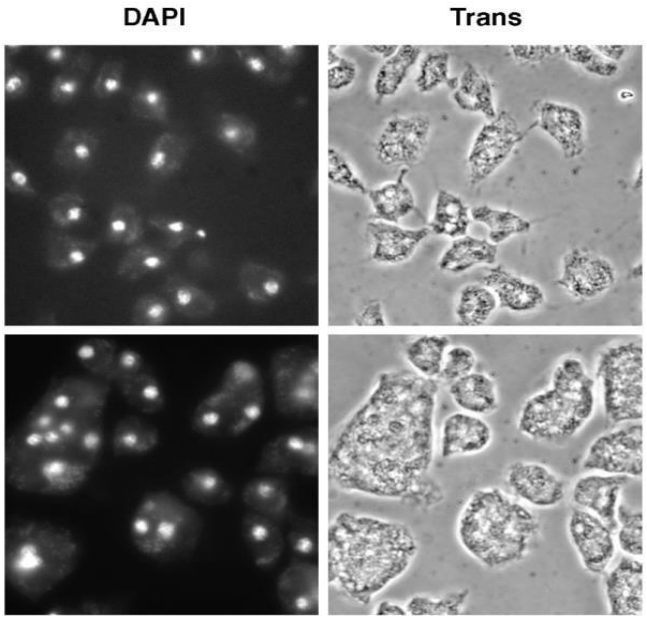


Figure 3. The top row shows the parental **AX3** strain and the bottom row is the **sadA-** mutant. On the left is DAPI staining the nuclei, and it shows how many sadA- cells are multinucleate. The transmission light image shows the size difference of the sadA- cells (bottom). In the panel of the AX3 cells (top), there is one dividing cell visible on the left side of the image. Note here cells are fixed, stained and under cover glass. The AX3 cells look fairly normal and comparable to cells in HL5. In HL5 the sadA- cells float but grow normally on bacterial lawn.

NOTE: *Add antibiotics to HL5 medium for mutants containing resistance to selective drugs on day 2. Prepare 50 - 100 mL batches of HL5 + blasticidin and HL5 + neomycin to replace plain HL5 in the process, these are the most common resistances.*

NOTE: *Not all mutant cells attach! For example, the sadA- mutant does not attach in HL5, instead becomes multinucleate and thus irregular large, floating but living cells. Cells like these need to be spun and resuspended gently in fresh HL5 to change the medium (see Fig. 3).*

Day 3 and Beyond: Maintaining/Shipping Axenic Cell Culture

1. Determine viability and density of the cells. Most viable cells adhere to the bottom of the plate, spreading into an irregular shape, and divide, cell density increasing over time. With non-attaching mutants estimate density to split or harvest cells. Dead cells float and appear very small and circular (do not confuse with living not attaching mutant cells). If cells appear as in picture 1 or 2 in ***Figure 3***, proceed to step 2. If cells appear like picture 3, allow to grow a few more days. If cells look as in picture 4, those are dead cells and need to be discarded.
2. To remove cells attached to the bottom of a plate, by gently pipetting the medium up and down numerous times over the whole surface to dislodge the cells. If the cells need to be split when too dense (picture 2) proceed to step 3. If cells are ready to be shipped (picture 1) proceed to step 4.

3. Prepare a new petri dish with 5-9 mL of fresh HL5,with antibiotic if applicable. Transfer 1-5 mL of HL5/cell mixture from the original plate to the new plate and discard the remaining cells. Gently swirl the plate to disperse cells.

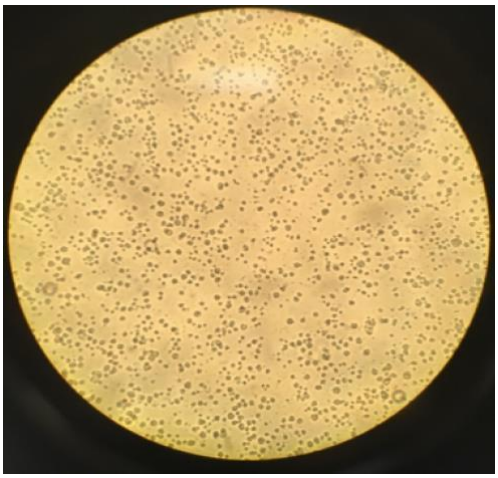
NOTE: *A dish generally holds 10 - 11 mL of HL5. The cell density is estimated and the volume of HL5 adjusted to dilute the cell suspension. Thus, if 3 ml of the cell suspension are used, prefill the new dish with 7 ml HL5, which is an approximately 1:3 dilution. Determine the cell volume to transfer based on how fast the cells grow, and when to be ready for shipment.*

4. For shipping, transfer the HL5/cell mixture to a labeled 15 mL conical tube.
5. Seal the tube with parafilm over the lid.

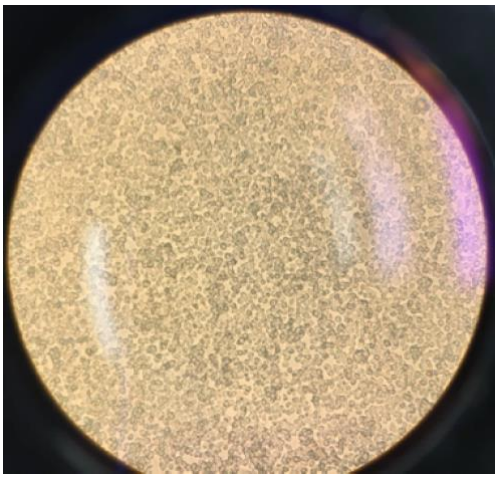
NOTE: *Cells in HL5 are fragile. Only ship the cells if the package will reach the user in 1-2 days. (e.g. do not ship HL5 cells on a Friday because it will not reach users before Monday). If cells cannot be shipped they need to be split to avoid overgrowth.*

Figure 4. Cells growing axenically at different densities

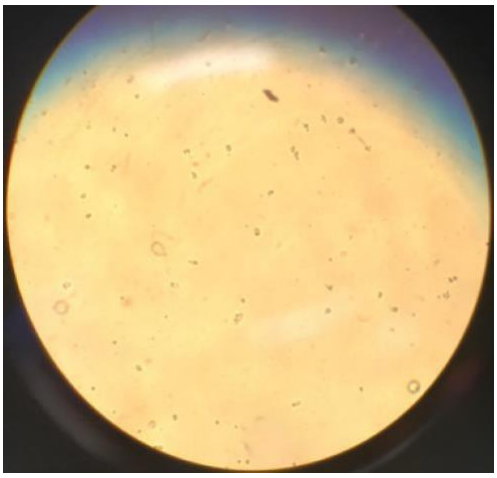
Picture 1: Viable cells at ideal density



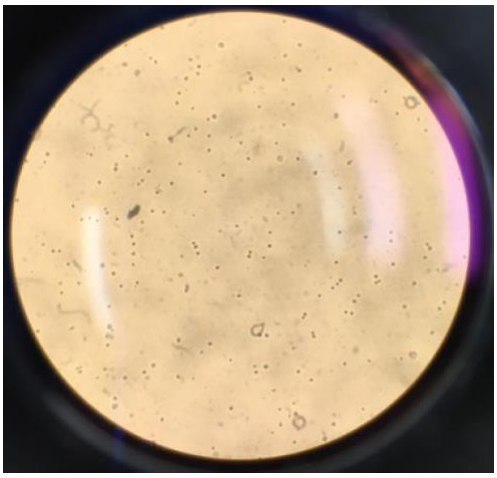
Picture 2: Viable cells at too high density



Picture 3: Viable cells at low density



Picture 4: Non-viable cells



To see how bacterially grown cells or developed spores are used to initiate a new axenic culture, refer to initiating axenic culture, page 14.

- Some axenic cells are adhesion mutants and do not attach.
- Adhesion mutants are often big in HL5 because they are multinucleate (see Fig.3). These cells might need to be spun down removing the smaller debris in repeated steps if necessary to enrich the live cells.

Axenic Growth in Suspension

1. Inoculate cells at about $5x10^4$ cells ml^{-1} in a glass flask containing HL5 medium, shaking at room temperature (20-22°C), 180 rpm. To achieve sufficient aeration.
2. the volume of medium should not exceed 20% of the volume of the flask.
3. Subculture after 2-3 days by making a 1:100 dilution into fresh medium, before the maximum cell density reaches $2x10^7$ cells ml^{-1}; it is not desirable to grow cells beyond $4x10^6$ ml^{-1}!

NOTE: *It's best to start a culture from cells established recently(!) in a petri dish; if cell density is below $2x10^4$ ml^{-1}, cells might go through a lag phase and have difficulties growing in suspension. In the DSC, this method is used for tests only.*

- ➢ Axenic medium easily contaminates, so all manipulations need to be done in the sterile hood.
- ➢ Axenic strains should be transferred to HL5 only when attendance is guaranteed on consecutive days for maintenance. At least 2 consecutive days are necessary to initiate the culture and then replace the medium. Ideally, HL5 cultures are started early in the week.
- ➢ Cells mutate easily when grown in HL5 for extended periods, especially when growing too dense. Do not maintain any strain for longer than 4 weeks in HL5 culture!

Sucrose Spores

Sporulating strains may be prepared as a sucrose spore pellet. Sucrose spores provide a durable and cheaper alternative for strain shipments to countries with extreme heat or strenuous customs policies.

Establish Strain

- Streak the strain on a bacterial plate and wait 2-4 days for healthy growth.

Grow and Harvest

- Add 650 µL of *E. coli*/Br in LB to a sterile microcentrifuge tube.
- Using a sterile loop, pick a large amount of cells from the feeding front of the original plate and inoculate the strain into the microcentrifuge tube (2 -3 loops might be used). Vortex gently to completely disperse the cells.
- Pipet 300 µL of the microcentrifuge tube content onto each of two GYP plates. Spread the contents on each plate evenly.
- Grow and develop the strain for 4-6 days, to mature fruiting body development over the entire plate.

Wash

- Using a sterile loop, gather spores from the harvest plates into a microcentrifuge tube containing
  1.5 mL of 10 mM KK2 buffer containing 30% sucrose
- Spin the tube at 12,000 rpm in the microcentrifuge for 1 minute then remove as much of the clear supernatant as possible without disturbing the pellet
  NOTE: *The spore material is sticky; swirl the loop in the buffer and / or tap to the side of the tube to release the spores.*

Dry, Store, Test

- Dry the pellet in the speed vacuum for 20-30 minutes
- Store the resulting honey-like pellet in the fridge for up to 3 weeks

- To resurrect the strain, inoculate a part of the pellet into HL5 or onto a bacterial plate using a sterile loop or toothpick.
- If resurrection is successful, the pellet can be shipped to the customer.

GWDI Mutants

GWDI strains are originally stored in 96-well plates with each well occupied by one or multiple mutants. The clones have been identified through the REMIseq method in which each insertion site has been sequenced. The 96-well plates are sealed by foil and stored at -80 ºC in Freezer B.

Overview:

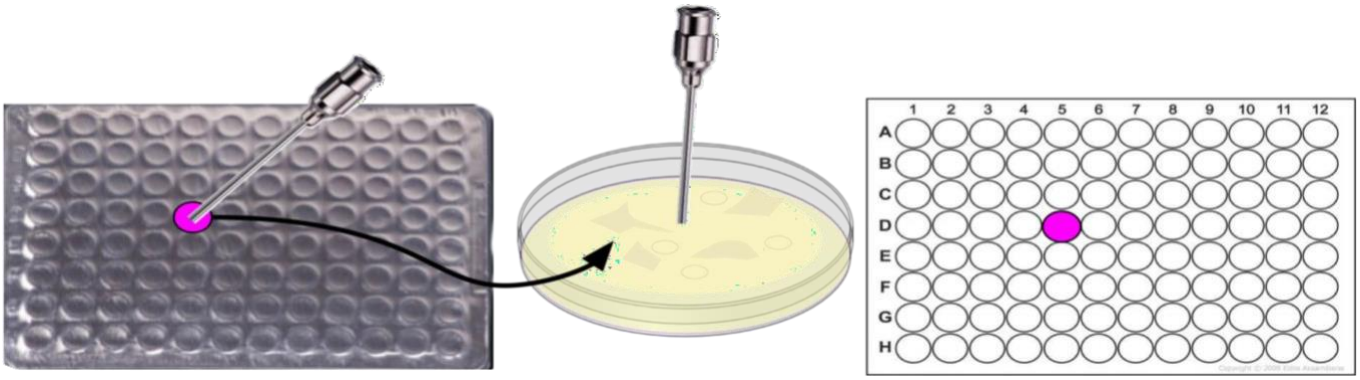


Figure 5. Overview of GWDI hatching process, with 96-well GWDI plate, needle, GYP plate and 96 well marking sheet

NOTE: *DO NOT open the well completely with first pick! Just poke a hole and scoop.*

Hatching the well

Preparation

- Wear gloves when handling sterile needle
- Prepare GYP plates with *E. coli*/Br
- Arrange Bunsen burner, icepack, sterile picking needles, bleach/water mix (about 30%) in a small container, aluminum foils squares (pre-cut), prepared GYP plate(s), and a blank 96-well-plate guide sheet to mark as you pick

GWDI hatching materials
GYP plate(s) with *E. coli*
Bunsen burner
Dry ice bucket
Icepack in bucket
Metal needle(s) sterile
Foil squares, pre-cut
GWDI marking sheet
30% Bleach/water mix

Table 4. Materials to prepare for GWDI plating

Retrieve Plate(s) from Freezer

- Mark plates to retrieve on GWDI Freezer Rack Maps..
- Get two blocks of dry ice for up to four 96-well-plates
  NOTE: *For 5 or more plates, use two dry ice buckets*

Just before Hatching/Picking

- Clearly mark the Letter & Number combo on the 96-well-plate marking sheet.
- Label each GYP plate with name (coordinates).

First Pick

- Place plate with clone(s) to harvest to the left of the ice pack.Use sterile tweezers to select a needle.

- Check 96-well-plate to make sure it matches marking on GYP plate.
- Firmly grip the needle (place the head of the needle between your index finger and middle finger)
- Puncture foil of well of interest all the way to the bottom and dig up contents.
  NOTE: *DO NOT completely open up well! Just poke, scoop, and break up cell chunk. This will ensure easy process of extracting well contents and easy re-sealing. Picking well multiple times to retrieve content is fine.*
- Quickly flip over the 96-well-plate, locate opened well over agar plate and tap plate to get the remaining frozen cell chunks out onto GYP.
- Repeat extraction process until well is entirely empty.
- Place 96-well-plate back in dry ice
  NOTE: *DO NOT warm up the plate. Work quickly and clean, otherwise other clones in plate might lose viability.*

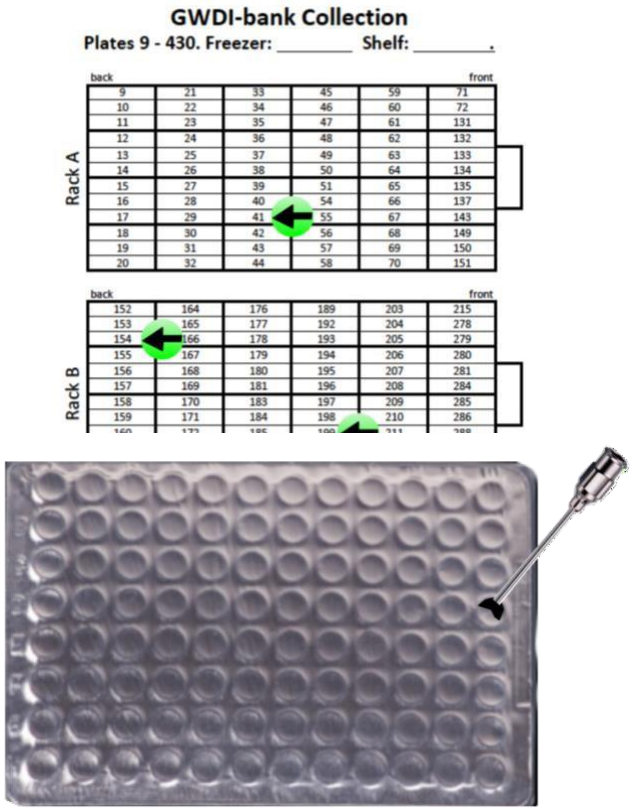


Figure 6. GWDI Freezer Rack Map marked, and First pick of well-of-interest

Finishing up

- Put used needle into bleach solution.
- Spray kimwipe with 80% ethanol and dab hatched well
  NOTE: *DO NOT wipe! Only dab opened well in order to avoid any cross contamination.*
- Peel and place pre-cut aluminum square on hatched well to cover and protect neighboring wells.

Repeat with each plate

- Use different colored sharpies for each new well hatched for easier identification.

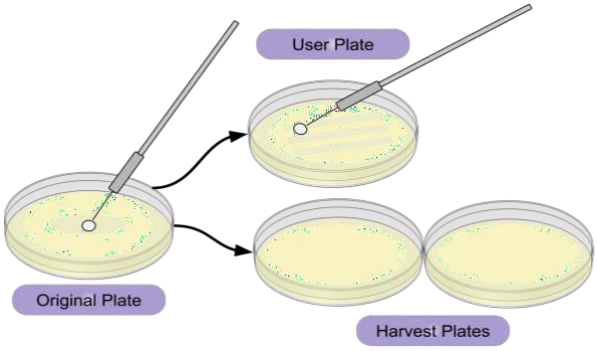


Figure 7. User and harvest plate process

Prepare User and Harvest Plates

1. After enough growth occurred on original plate, scrape cells from the feeding front of the original plate to a fresh bacterial plate labeled for the user.

2. Inoculate two harvest plates for strain storage the same way.

3. Store single GWDI mutants as spores if they sporulate, otherwise as cells.

Hatching a well with multiple clones

- Wells containing multiple mutants (mini-pools) need special attention.
- Harvest all growing cells not only the feeding front to make sure all mutant clones are represented.
- Mix well before plating a user plate and one or two harvest plates.
- Always store as cells not as spores, as one or more of the mutants might not fruit.

NOTE: *In a mix of different mutants, one might have slower growth rate and another might not fruit. Therefore, make sure to take all cells from the original plate to proceed onto harvest plates.*

Plating and Growing Plasmids

The plasmid ID is a unique identifier. Plasmid names are entered as described or published; only if a lower case 'p' is missing, this will be added and the first letter capitalized, e.g. arpB-KO will become pArpB-KO and the original name is added as synonym.

Plasmid information and storage

ID	Name	Stored as	Location	Color
DBP0000203	pDXA-GST	in DH5α	4(45,54)	pink

Figure 8. Sample plasmid order

Plasmids are available as a DNA sample and/or bacterial transformant for direct replication in a bacterial host such as XL10 Gold, DB3.1, or DH5alpha, an example is seen in Figure 9. For both forms of plasmids, the first number indicates the box number, and the number in parentheses indicates the exact position(s) within the box. Unless otherwise indicated by the user, plasmids are shipped as bacterial transformant.

NOTE: *Check the plasmid record! Nearly all plasmids carry an ampicillin resistance and are plated on LB/Amp plates. However, a plasmid may be kanamycin resistant and thus must be plated on LB/Kanamycin plates.*

Growth of Bacterial Transformant

1. Label an LB/Amp plate with the plasmid ID, name, and date.
2. Using an ethanol-flamed sterile screwdriver, streak bacteria from the stock tube in section 1 (Fig. 10).
3. Using a sterilized loop, streak bacteria into section 2, as shown by the green line.
4. Sterilize the loop again, and streak bacteria into section 3, as shown by the yellow line, to receive single colonies. You should get more single colonies than shown on the picture.
5. Incubate at 37ºC overnight (or at room temperature over 2 days).
6. Parafilm seal the plate and ship immediately or store in fridge short term before shipment.



Figure 9. Streaking plasmids to obtain single colonies.

Preparation of DNA on Filter Paper

NOTE: *Wear gloves throughout this procedure, your hands may contain nucleases or other degrading or contaminating proteins.*

1. Spread a clean sheet of Saran wrap on the clean bench.
2. Place a small piece of white filter paper in center of saran wrap.
3. Using a pencil or pen, write the name of the plasmid at the bottom of the filter paper and draw a 5-8 mm diameter circle in the center.
4. Spot 1-2 microliters of DNA onto center of the drawn circle, and wait a few minutes for spot to dry before wrapping it in the saran wrap.

Shipping Strains and Plasmids

Shipping Orders

Ideally regular orders are shipped within 10 days of being placed. GWDI strains usually take at least two weeks until they are ready to be shipped. If an order is taking longer, make sure to email the user explaining the reason for the delay. Pure plasmid orders should typically take no longer than 2-4 days, depending on when in the week orders are placed.

GITHUB

After shipping an order, remove the Growing / in Preparation label and add the Shipped label; add the ship date and tracking ID in the comments section.

We strive to ship all items in an order together. However, when we have large orders, it is at our discretion to send multiple shipments and inform the user.

Bacterial plates (*Dictyostelium* or plasmids)

1. Wipe off any condensation formed in lid using a kimwipe while wearing gloves.
2. Parafilm seal the plates.
3. Up to 3 plates fit into one cardboard box. Use cotton absorbent towels to cushion and protect the plates.
4. Place box into a FedEx/UPS/DHL pak with appropriate shipping label.

HL5 Cultures

1. Label a 15 mL conical vial with DBS ID, name, date, HL5 drug concentration used
2. Transfer 10 mL of HL5+cells into the labeled 15 mL tube.
3. Parafilm the top of the vial.
4. 3-4 tubes fit into one box; follow steps 3-4 above.

Supplies

- Cardboard boxes are ordered from https://www.packagingsupplies.com/brown_kraft_1-piece_folding_cartons.html (product ID C342H6).
- Cotton towels are ordered from https://www.fishersci.com/shop/products/cep-biodegradeable-sorbents-7/191500196?keyword=true (catalog # 19-150-0196).
- FedEx and UPS shipping supplies are ordered for free from their respective websites. DHL shipping material may be obtained by requesting supplies from DHL when you schedule a pickup or ask Michelle if there are DHL envelopes available.

Shipping Documents

1. Account Number/ Premade Label

Users must pay for shipping by giving an account number or by emailing dictystocks a shipping label. **NOTE:** *FedEx and UPS labels can be created online. DHL labels are created using DHL Express eMailship.*

2. Label Information (Domestic / International)

- Shipper Address
  Olivia Martin Northwestern University 303 E. Superior St Simpson Querrey 5-500
  Chicago, IL 60611 United States
  312-503-4169
- Product Information
  Cultures of *Dictyostelium discoideum* and/or bacterial plasmids. Non-toxic. Non-pathogenic. Research use only.
- Weight/Declared Value 1 lb/$10
- Packaging Type
  FedEx Pak UPS Pak DHL Pak
- Service Type
  FedEx Priority Overnight/International Priority UPS Next Day Air / Worldwide Saver (Express)
  DHL Express 12:00
- Payment Info
  Account number

NOTE: If the DSC is paying, make sure to add our current chart string to 'Your reference' section

3. Recipient International Customs Forms

International orders require three customs forms: declaration of biological shipments (instructions on filling out this form shown below), manufacturer's declaration, and 4 copies of a commercial invoice. FedEx and UPS will generate a commercial invoice when checking the box for commercial invoice. A manually filled out commercial invoice is only needed when shipping through DHL or creating a manual shipping label for FedEx.

4. Declaration of Biological Shipments Information

- Reason for export
  Check "Sample for lab research"
- The above shipment contains:
  Write N/A over sections for "Human", "Animal, "Biological Specimen", and "Diagnostic Specimen"
  Check "Other"
  Dicty description: *Harmless samples of Dictyostelium discoideum that are non-toxic, non- pathogenic and with no animal or human origin. For research use only.*
  Plasmid description: *Harmless bacterial plasmids for research use only. Nonpathogenic, nontoxic, with no animal or human origin. Harmonized code: 300290500000*
- Parts of animal bodies and source material (plant or animal name): Write N/A over these boxes
- Description of the packaging
  Indicate what materials are being shipped in (ex. FedEx/UPS/DHL pak, Styrofoam box, cardboard box, etc.)
- Emergency Contact/ Declarants Name
- Enter your information

NOTE: *If you are shipping antibodies (see Shipping Orders- Shipping Antibodies and DNA Libraries, page 20), check "Biological Specimen" and write N/A over every other section. In the "Biological Specimen" section, check the animal the antibody was derived from.*

5. Drop Off Package

FedEx: Drop off packages in the Northwestern Mailroom (located in basement of Morton Medical Research Building) before pick up at 3 pm.

UPS: Drop off packages at the UPS store 207 E Ohio St, Chicago, IL 60611 before pick up at 6 pm. Another UPS store is on 40 E Chicago Ave, Chicago, IL 60611, before pickup at 6:30 pm.

DHL: Via DHL Express eMailShip', select schedule pickup. The DHL driver will call you when they arrive either at the SQ lobby or at the SQ delivery dock.
https://www.dhl.com/en/express/resource_center/emailship.html or Call 1-800-225-5345 to schedule pick up.

6. Email User

Use the following format to notify users that their order has been shipped:

> Dear Dr. [**insert their name**],
>
> I shipped your Dicty Stock Order today via [insert shipping carrier]. The tracking ID is [insert tracking ID]. Please let me know when you receive your order and if you have any questions.
>
> Kind regards,
> [**insert your name**]

7. Track the Package

Use the carrier website to track packages. Once the order has been delivered, update the order status in GitHub by adding a comment indicating the order has been delivered, with tracking details. The order is now ready to billed.

GITHUB
When the order has been delivered, add the delivery details in the comments section.

Shipping Sucrose Spores

For shipments made out to countries with hot climates or ones with frequent customs delay, sporulating strains can be sent as sucrose spores (see Plating Orders - Sucrose Spores, page 17. To pack, take the dried sucrose spore pellet out of its tube and wrap it in two air-tight layers of saran wrap, flatten gently and ship as a regular letter as described below in the "Shipping Plasmids on Filter Paper" section.

Shipping Plasmids on Filter Paper

For orders that consist of plasmids only, we may ship plasmids on filter paper via USPS as a letter at our discretion. No more than 5 plasmids may be shipped in a single letter. Place the wrapped filter paper(s) into an envelope with either a domestic or international Forever Stamp purchased for us by Michelle Mohney. These orders cannot be tracked, thus make sure to follow up with the user on the delivery status of the order. This option is mostly used by international custo mers to save on shipping and with expected custom difficulties.

Shipping 'Additional Materials'

Users also order antibodies, DNA libraries, and GWDI pooled libraries through the DSC. These items do not require any "growing / in preparation" stage and can be shipped readily. It is important that these materials stay frozen during shipment, thus they must be shipped with plenty of dry ice, packaged within a Styrofoam box, which is contained in a cardboard box. Refer to FedEx's Dry Ice Aid.

Before preparing an international shipment, make sure to call FedEx, 1.800.463.3339, to check if the country accepts dangerous goods like dry ice. E. g. Turkey, as of 2019, does not accept international dry ice shipments. Sometimes, a DHL account is also not setup



Figure 9. FedEx representation of how-to shipment with

to accept dangerous goods such as dry ice, so ask user to verify this clearance for their account number.

The shipping label must indicate that the package contains dry ice, and the weight needs to be modified to reflect the amount of dry ice. A dry ice sticker must also be applied to the box.

Modify the commodity description on the custom form for antibodies and DNA libraries. Additionally, for antibodies, for the declaration of biological shipments, check the "Biological Specimen" section and indicate the animal the antibody was derived from. For GWDI pools (mutant library), the description of the materials can be the same one for when Dicty is shipped on plates.

Strain and plasmid deposits

1. Make sure to have the user fill out a strain or plasmid deposit form and email to dictystocks email
    a. https://dictycr.org/stockcenter/information/deposit
2. Give the user our DSC FedEx account or create a label and send to the user.
3. After safely storing the strain or plasmid, add strain or plasmid to Excel file tracking deposits
4. Ask plasmid submitter for plasmid map (restriction sites) and / or sequence.
5. Follow Data Curation guidelines (page 38) and add strain or plasmid to the database with the appropriate annotations

Payment Procedures

As mandated by our grant, users are required to pay a small fee for the services associated with the materials they order from the DSC. All charges go through NUcore, Northwestern University's billing system. The prices are displayed in the table below:

Price Group	Strains	Plasmids/Food Strains	Other Materials*	GWDI Pools
Non-Profit	$30	$15	$40	$100
For Profit	$100	$50	$200	$400
Waiver	$0	$0	$0	Not applicable

Table 5. NUcore material prices
**Other Materials: Antibodies and DNA libraries*

Most users fall under the non-profit price group. We rarely have commercial users, who need to write that they are not using the materials to make any profit or sell it to others. Some users qualify to have the charges for their materials waived, for example teachers who do not have funds for Dicty strains.

It is important to distinguish between the cost of the **materials** and the cost of **shipping**. Users are responsible for both. We only cover shipping costs if we made an error in an order, or if a lab is submitting materials to the DSC (see page 24 and Storage Procedures, page 25-29).

Updating prices in NUcore

1. Go to 'Products
2. Select 'Services' - All materials are considered an 'active service
3. Select a service: GWDI pooled library, Other material, Plasmid/Bacterial Food Strains, Strain
4. Select 'Pricing'
5. Reset the effective dates for the next year.
6. Repeat process for each of the four services

NOTE: *Need to confirm price each fiscal year around September 1.*

Verifying Payment Information

Verify the following pieces of payment information have been entered correctly on order forms:

1. Organization/Lab: Users from the same laboratory usually have the same payment means. Even users in the same department might share the payer info. Therefore, if other users from the same department or lab place orders in the future, the payer information may already exist in NU Core.
2. Payer Name/Email: In order to proceed with billing, we must create a user account in NU Core using the payer's name and email address.
3. Payment Method: Credit Card Purchase Order Wire Transfer Chart String Payment (internal NWU customers)

NOTE: *Customers who pay by wire transfer need the Northwestern bank information emailed when orders are shipped. Bank info files differ for national and international customers, sent accordingly.*

GITHUB

Add the specific label for payment method



Add a NEW User to NUcore

1. Go to https://nucore.northwestern.edu/facilities/Dicty and log in.
2. Click “Manage Facilities” in the upper right-hand corner and select Dicty Stock Center.
3. Click “Users” in the top and enter the payer’s email in the search box.
4. If the user is new, the search will be unable to detect anything; click “Create a new user.”
5. The payer does not have a NetID because they are not from Northwestern, reenter their email.
6. No information on the payer will be found in NUcore, click “Add a new external user.”
7. Enter their name correctly because names cannot be edited later, then click “Create.”

Adding the Payment Source to NUcore

Once a user has been added to NUcore, he/she needs to be linked to a payment source.

1. Click on “Payment Sources” from the menu on the left-hand side of the page.
2. Search the user by their email address in the search bar, and no payment sources for the user should be found.
3. Click “Add Payment Source,” then reenter the user’s email in the search bar when prompted.
4. Click on the user’s name to proceed to the list of payment options.

Credit card payment

1. Select “Credit Card” from the tabs on top.
2. Enter name of credit card owner, which ideally is the payer/user, but may be different.
3. For the expiration month/year choose any date in the distant future.
4. For the description, enter **[Payer Name/Lab/CC Account]**.
5. For the affiliate choose “other,” and enter the university/organization the payer is associated with.
6. In “Bill To” enter the full address of the payer.

NOTE: Users who pay by credit card are charged a 3.5% bank fee, which is not reflected in the DSC order invoice, but they see it when paying online. Explain to users as necessary.

Purchase order payment

1. Select the “Purchase Order” tab.
2. For the account number, enter the PO number obtained.
3. If an expiration date has been provided, enter. If not, choose any date in the future.
4. For the description, enter **[Payer Name/Lab/PO Account]**.
5. Continue with 5 of Credit card payments, above.

➢ You may view and edit a user’s payment sources at any time in case they wish to switch to a new payer and/or payment source.

➢ Payment sources for existing users will eventually expire upon the future date assigned. If the payer for a lab group stays the same but their payment source has expired, just recreate the payment source for the payer using the steps listed above.

Wire transfer

1. Select the “Purchase Order” tab (there is no wire transfer but this is ok).
2. For the account number, enter XXXX.
3. For the expiration date choose any date in the distant future.
4. For description, enter **[Payer Name/Lab/Wire Transfer]**.
5. Continue with 5 of Credit card payments, above.

Adding an Order to NUcore

GITHUB

After adding the order to NUcore, add the in NU Core label.

1. Search the payer in the Users tab.
2. Click the Order For, and then Order For. Enter in the number of each item ordered and create the order.
3. Select the appropriate payment source associated with the payer and continue.
4. For the note enter, as it shows in GitHub, [Order ID:#### USER NAME], then click "purchase."
5. Click "Exit" at the top of the page in the blue bar.
6. Click the Orders tab in the tool bar.
7. Click the check mark box to select the order, and change the Status markdown to Complete (the markdown is next to the 'Update' button)
8. Click on the "Billing" tab on the menu at the top of the page, which will lead to a page that says "Send Notifications."
9. Check the order and send the notification(s). The payer has now been notified that an order has been created in NU Core. This is a period in which the user may review the order, but they cannot yet pay.
10. An order may stay in the review period for 7 days; however, we typically mark orders reviewed a day or two after sending notifications. To do this, click on the "Orders in Review," check the order, and click "Mark as Reviewed." Most users do not review their orders.
11. Click on the "Create Statements" tab, check the order, and click "Create." An invoice in pdf format has now been sent to the user. For those paying by credit card, the statement includes a link that directs them to a secure site for payment.

Waiver Price Group

For individuals requesting waivers, orders are added to NU Core in the same manner as listed above. However, they need to be assigned to a new price group. To do this, follow these steps:

1. After adding the user to NU Core, click on "Admin" at the top of page and then "Price Groups" on the left- hand menu of the page.
2. Click "External- Waiver Approved" and then "Users."
3. Search the user you wish to add to this price group and add.
4. For Payment source, enter as if it were a credit card. Description write "Waiver."
5. Add the user's order AFTER assigning them to the waiver price group.

- An order may be canceled in NU Core if entered incorrectly or if there are duplicates of an order. To do so change the order's status from "New" or "Complete" to "Canceled." Once an order has been cancelled, it cannot be reversed; if cancelled by accident, you will need to recreate the order.
- Payment receipt: accessible by clicking 'Payment Link' after user has paid and screenshot can be taken. We or the user may access the link.

Once a user has been added to the waiver price group, they will stay until manually removed.
NOTE: *Waivers are approved by the DSC manager in direct communication with the user.*

No Charge Orders

No Charge is used when a user will not be charged for their order due to an error on our account. These orders do not need to be added to NU Core, and thus bypass the payment status steps in GitHub. They can be closed in GitHub once they have been shipped. "No charge" orders are not to be confused with "waiver/no payment" orders.

For Profit Price Group

For users from for-profit companies follow the same steps as above and choose appropriate price tags for this group, e.g.: FOR PROFIT – Strain. Selecting the "FOR PROFIT" material will list the items with the for-profit price on the invoice.

GITHUB

After sending the invoice to the user, add the Billed label and close issue.

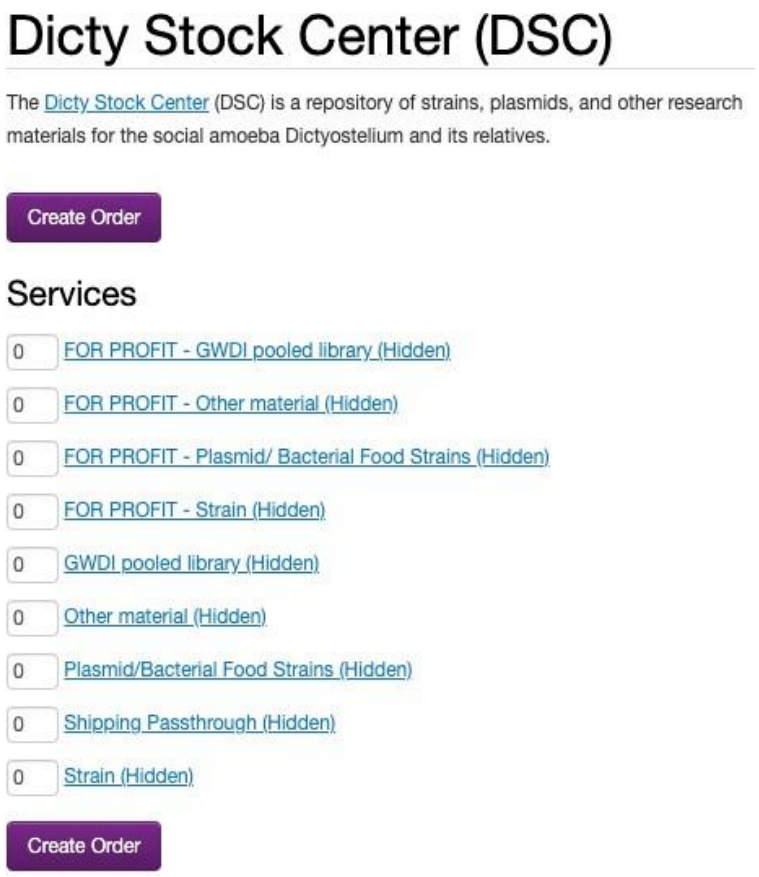


Figure 10. Overview of DSC material Price Tags

Storing Strains

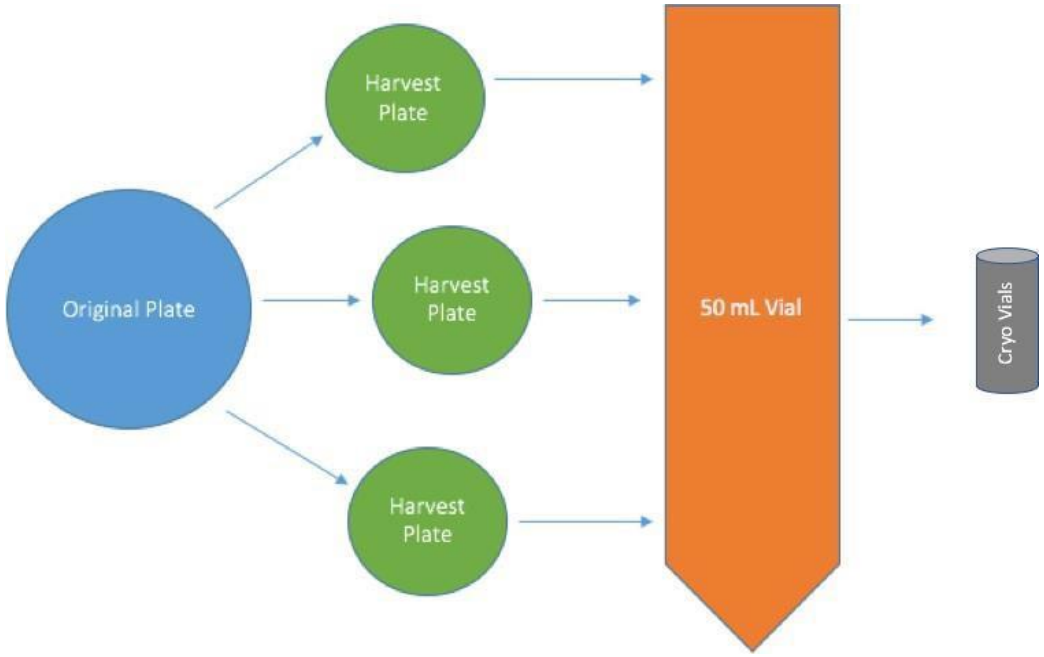


Figure 11. Overview of storage process

During cell storage, we establish several harvest plates of the same strain growing at the same density and/or to the same developmental stage. Wild type and certain mutant strains that form healthy spores, are often stored as **spores**. Cells that do not sporulate are stored as bacterially grown **cells**. All axenically growing cells may be stored **axenically**. We plan to offer users the choice of choosing a bacterial plate or cells in HL5, so it is ideal to have both spore/cell and axenic stocks for these strains.

During all procedures, cells growing on several harvest plates are combined into a single 50 mL tube, in which the cells are pelleted and then washed until unwanted media, buffer, or bacteria are removed. The concentrated cells at the end are then supplemented with freezing agent and split into a number of cryo- vials.

Storage from HL5

Establish strain

- Grow the strain to be stored in 10 mL HL5 for 2-4 days (original plate).

Harvest

- Set up 5 new plates with 8-9 mL HL5 (harvest plates).
- Dislodge cells from the original plate by pipetting media up and down several times. To detach cells, move over whole plate.
- Transfer 1-2 mL cell+HL5 mixture from original plate to each of the 5 new plates. Use the same amount for each plate!
- Gently swirl harvest plates to make sure cells are evenly dispersed.
- Grow cells to ideal density (see Plating Orders - Figure 4, Picture 1, page 16.

Centrifuge Speed
Centrifuge - Thermo Scientific Sorvall ST 8
- ➢ Cells: 1550 RPM
- ➢ Spores: 2500 RPM

Combine

- Combine the cells from the 5 harvest plates into one sterile 50 mL conical tube. Dislodge all cells by pipetting up and down and transfer all contents into the vial.
- After combining the contents of all plates, invert the tube a few times to mix well.

Count

- Set up a hemocytometer, such that the glass coverslip is securely affixed with moisture.
- Pipet 50 µL sample from the conical tube into one of the hemocytometer chambers.
- Count five of the 0.04 mm2 squares. For example, count the cells in the four corner squares and the center square (blue squares).
- Calculate the sum of cells counted in the five squares.
- Multiply the cell count by 50,000 to obtain the cell density (# cells/1 mL).
- Multiply the number of cells by the total volume (mL) in the conical tube to determine the total number of cells.
- Determine how many cryo-vials are needed, such that each vial contains approximately 5 x 10^7 cells per mL. The ideal yield is 10 vials (1 to test, 9 to split between the 3 tanks). Label cryo-vials with strain name and storage date and pre-cool empty tubes on ice.

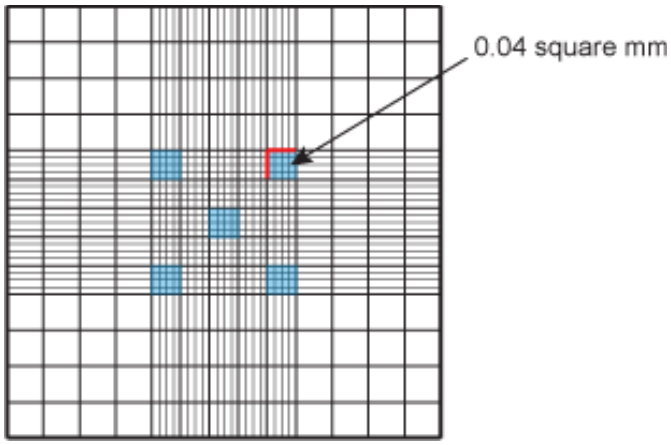


Figure 12. Hemocytometer setup for counting cells

Concentrate

- Centrifuge the cells at 1550 rpm for 4 minutes.
- Decant liquid calmly but quickly without disturbing the cell pellet at the bottom of the vial.

Split

- Place the vial containing the cell pellet on ice.
- Add the appropriate amount of fresh, cold HL5 to the pellet (e.g. to make 10 tubes, add 10 mL HL5) and completely resuspend the pellet.
- Add DMSO to 5% of the final volume (e.g. to 10 mL HL5, add 500 µL DMSO) and gently swirl to mix. Keep cold!
- Transfer 1 mL of the sample into each cold cryo-vial. Close the vials and keep on ice.

NOTE: *Cells are very fragile in the presence of DMSO. Use care and handle gently after adding DMSO. Do not vortex or vigorously pipet and keep the cells cold at all times.*

Freeze

- Freeze cells initially at 20°C, for 1-2 hours.
- Transfer the cells to the -80°C freezer, where they stay until 1 tube has been tested.
- After cell viability has been tested, distribute the remaining cryo-vials among the 3 liquid nitrogen tanks and add the new location into the inventory database for this strain.

Storage from Bacterial Plates - Cells

Establish Strain

- Streak the new strain onto a bacterial plate (original plate) and incubate for 2-4 days.

Harvest

- Add 650 µL of E. coli/Br in LB to a sterile microcentrifuge tube.
- Using a sterile loop, pick a large amount of cells from the feeding front of the original plate to inoculate the strain into the microcentrifuge tube (2-3 loops might be used). Vortex to disperse the cells evenly.
- Pipet 300 µL of the microcentrifuge tube content onto each of two GYP plates. Spread the contents to evenly disperse on both plates.
- Allow the strain to grow for 2-3 days, or until the plate looks more opaque and most bacteria have been consumed.

NOTE: *It is important to store healthy, vegetative cells that have consumed the vast majority of the bacteria, but are not yet starving, initiating development. The strain is ready when the smell of the plate changes from a bacterial odor to the scent of Dictyostelium (woody/earthy). If harvesting occurs too early, cell count is low and washes show heavy bacteria content. If little bumps appear, development started leading to gene expression changes, and cells cannot be stored. Non-fruiting cells still starve, those must be harvested at the same critical time point when storing as cells.*

Combine

- Add 4 mL of 1X KK2 buffer to each harvest plate.
- Use a sterilized glass spreader to detach cells from the plate, then transfer strain/buffer mixture to a 50 mL conical tube.
- Repeat this step with 2 mL of 1X KK2 to make sure all cells have been transferred to the vial.
- After transferring all cells, fill the tube with 1X KK2 buffer to the 50 mL mark. Invert the tube several times to mix.

Wash

- Centrifuge cells at 1500 rpm for 4-5 minutes. During this process, the larger and heavier *Dictyostelium* cells will form a pellet at the bottom. The lighter bacteria will remain suspended in the buffer.
- Decant liquid quickly but gently without disturbing the pellet at the bottom.

Count

- Resuspend cell pellet in 50 mL 1X KK2 buffer. Make sure cells are evenly distributed!
- Follow counting procedures detailed in the axenic storage (Storing Strains- *Figure* 11 above).

Wash

- Repeat wash as in the previous wash step.
- If the liquid to decant appears clear, proceed to next step. If still cloudy, indicating the presence of remaining bacteria, wash again (3 washes total).

Split

- Label at least 4 cryo vials (1 to test, 3 to split between the 3 tanks), depending on the number of cells. Pre-cool vials on ice and label with strain name and storage date.
- Place the 50 mL tube containing the pellet on ice.
- Add the appropriate amount of fresh, cold HL5 to the pellet (e.g. to fill 4 tubes, add 4 mL HL5) and resuspend the pellet.
- Add DMSO such that the final concentration is 5% (e.g. to 4 mL HL5, add 200 µL DMSO), then gently swirl. Remember to be gentle after addition of DMSO!
- Distribute 1 mL sample from the 50 mL conical tube into each pre-cooled, labeled cryo-vial.

Freeze

- Freeze cells initially at 20°C for 1-2 hours.
- Transfer the cells to the -80°C freezer, where they stay until tested.
- After the cell viability has been tested, transfer the remaining cryo-vials to the 3 liquid nitrogen tanks and add the new location into the inventory database for this strain.

Storage from Bacterial Plates- Spores

Establish & Harvest

- Establish and harvest the strain as done for cell storage, above
- Allow strain to grow on 2 harvest plates for 4-6 days, to develop mature fruiting bodies.

Combine

- Use a sterile loop to transfer allfruitingbodies/spores frombothplates to a 50 mL conical vial, filled with 30 mL KK2 buffer
- After transferring all the spores, fill the tube to 50 mL with 1X KK2 buffer. Invert the tube several times to distribute the spores evenly.
  NOTE: *Stalksare pickedup when transferring spores to thetube,but these are unviable, vacuolated support cells. However, the material is sticky and cannot be counted, as done for axenic or bacterial cell storage. We found that two plates of fruiting bodies result in enough spores to split between four tubes.*

Wash

- Centrifuge spores at 1650 rpm for 4-5 minutes
- Decant liquid quickly but gently without disturbing the pellet at the bottom
  NOTE: *Since spores are transferred directly to a tube containing very little bacteria, they only need to be washed once. Also, spores are the most robust storing method, if available.*

Split

- Label at least 4 cryo-vials (1 to test, 3 to split between the 3 tanks. If >4, store 2 vials per tank. Pre- cool vials on ice and label with strain name and storage date.
- Place the 50 mL tube containing the pellet on ice.
- Add the appropriate amount of fresh, cold HL5 to the pellet (e.g. to fill 4 tubes, add 4 mL HL5) and resuspend the pellet. To disrupt spore clumps pipetting up and down gently, helps to disperse.
- Add DMSO such that the final concentration is 5% (e.g. to 4 mL HL5, add 200 µL DMSO), then gently swirl. Remember to be gentle after addition of DMSO!
- Distribute 1 mL sample from the 50 mL conical tube into each cold, labeled cryo-vial.

Freeze

- Freeze cells initially at 20°C for 1-2 hours.
- Transfer the strains to the -80°C freezer, where they stay until tested.

After the cell viability has been tested, transfer the remaining cryo-vials to the 3 liquid nitrogen tanks and add the new location into the inventory database for this strain.

Troubleshooting – Phenotype and Cloning

Occasionally, a user will report a discrepancy with the phenotype of a strain received. With any phenotype issue, we first need to consider the published methods used to culture the strain. SM or SM/5 pl ates are typically used in research studies. SM plates promote much more bacterial growth than LPB, and the development of some mutants can be stalled at certain points of development. Typically use GYP plates when phenotypes were reported on SM or SM/5 as it's richer than LPB. For culturing Dicty axenically, the described antibiotic concentrations are very important to review if there are issues with cell survival.

Cloning

The cloning method can be used to evaluate phenotype discrepancies by selecting individual cell colonies in order to make better observations. The following instructions are to be completed for both the strain in question and a control strain.

1. Plate the strain in question on GYP and sometimes on LPB as well, and a control strain and wait for growth
    a. Control strains can be AX2, AX3, AX4, NC4.
2. Prepare conical tubes per plate with 5 mL of 1x KK2 buffer
3. Prepare 12 LPB and/or GYP plates, depending on methods described to prepare strain in question
4. Wait for a sufficient feeding front and with a sterile loop, pick the feeding front and place into respective tubes
5. Vortex conical tubes
6. Count the number of cells in the sample
    a. Refer to 'Storage from HL5' instructions how to count cells
7. Dilute the cell solution so that the cell concentration is between 10 – 50 cells per plate.
8. Calculate the number of bacteria needed and add the appropriate amount of cell solution to microcentrifuge tube
    a. 1 cell counted (50,000)(5 mL KK2) = 250,000 cells/mL
    = 250 cells/µL M1 x V1 = M2 x V2
    250 cells/µL x V1 = 25 cells x 250 µL
    V1 = 20 µL cell solution per plate + 230 µL bacteria
9. Add new solution of cells and bacteria to each plate and spread with sterile spreader
10. Observe plates for clones or plaques in bacterial lawn
11. Pick clones onto new plates for further phenotype analysis

Storing Plasmids

Storage as Bacterial Transformants

1. Pick a single transformed colony from an LB plate and inoculate into 2 mL LB broth + drug (usually ampicillin) in a round bottom 10 mL tube.
2. Grow the culture in a shaking incubator at 37°C overnight.
3. Start a new culture for storage by transferring 0.2 mL of overnight culture to a new round bottom tube containing 3.8 mL LB broth + drug.
4. Grow culture in shaking incubator at 37°C for 4 hours.
5. Cool the culture on ice for 10 minutes or more.
6. Add 1 mL of cold 75% glycerol for a 15% final concentration and vortex gently.
7. Add 1 mL of sample to each of 4 cold cryo-vials, prelabeled with the plasmid name and date.
8. Store and freeze the plasmid in the -80°C freezer and add new plasmid inventory into the database.

Storage as DNA

To store plasmid DNA, aliquot the sample and freeze at -20°C. Enter the location into the inventory database.

Molecular Procedures

Dictyostelium expression plasmids are submitted either as (A) DNA spotted and dried on filter paper, (B) DNA solution in a tube, or (C) transformed bacterial strain growing on an agar plate.

Plasmids received on filter paper contain traces of DNA that must be eluted off the filter paper to create a low concentration DNA solution. Eluted filter paper plasmids or plasmids received directly as DNA solution are used to transform electrocompetent bacterial cells. A miniprep, restriction digest, and gel electrophoresis are performed with at least two independent transformed clones to verify successful transformation. Handling DNAs and all molecular procedures should be carried out very cleanly and wearing gloves.

Elution

1. Aseptically (wear gloves!) cut out the paper circle and transfer to a microcentrifuge tube
2. Add 50 µL TE buffer and push paper gently to bottom of tube using the pipette tip so it is in contact with buffer
3. Vortex the tube briefly
4. Perform a quick spin to collect the solution at the bottom of tube
5. Remove the paper from the tube using forceps (ethanol cleaned) and save in case the extraction failed and needs to be repeated.
6. Perform transformation with 2 µL eluted DNA

Electrocompetent Cells

Electrocompetent cells in the DSC are typically made from the XL10 Gold strain and used for the transformation process. Prepared electrocompetent cells can be found in a labeled box in the -80°C freezer A, shelf 3. Follow the procedure below to prepare electrocompetent cells.

Materials

- 1 LB Plate (no antibiotics)
- 1L LB broth (no antibiotics)
- Two autoclaved 2L flasks
- 2L ice cold autoclaved DI water (electrocompetent water)
- Two 250 mL autoclaved centrifuge bottles

- 1 sterile 50 mL tube
- 100 mL autoclaved, ice cold, 10% glycerol
- 50+ sterile microcentrifuge tubes

NOTE: *All reusable plastic and glass materials used for this procedure must be rinsed out with deionized water several times to get rid of all soap residues before sterilization. Ideally, the materials used to make electrocompetent cells are reserved for this procedure only.*

Protocol

1. Streak the XL10 Gold cells onto a plain LB plate and grow overnight at 37°C
2. Inoculate a single colony into 25 mL of LB in a 125 mL flask, grow shaking overnight at 37°C
3. Transfer 2.5 mL of overnight culture into two sterile 2L flasks each with 400 mL LB
   NOTE: *This is the ideal volume to grow the cells to prevent them from getting too dense too quickly.*
4. Put 2L flasks in the shaking incubator and let grow at 37°C (typically for 3-4 hours)
5. Check the OD of the culture every hour until an OD of 0.5-0.7 is achieved
   NOTE: *To check the OD, add 1 mL LB to a cuvette and press the cuvette button. This is the blank and the reading should be 0.00. (Blank every time after the machine was off). Add 1 mL sample into a new cuvette and press the green button. The number is the OD reading.*
6. When desired OD is reached, chill the flasks in an ice-water bath for 15+ minutes.
7. Transfer the cells to 2 cold, autoclaved 250 mL centrifuge tubes. To discard excess cells (~450 mL), add bleach and let the cells sit for at least 30 min. before rinsing with plenty of water.
8. Spin cells for 15 min at 4000 rpm/ 4°C in the Avanti centrifuge
   NOTE: *Avanti centrifuge is owned by the McNally lab and located in SQ 5-419. If you need help, just ask a McNally lab member!*
9. Pour off supernatant quickly; pellet is relatively loose at this stage.
10. Wash cells with ice cold electrocompetent water. To wash add 3 increments of 25 mL water, swirling each time. When pellet is in solution fill the tube with water.
11. Repeat step 8
12. Repeat step 10 but combine the two pellets into one tube before filling the tube with water.
13. Repeat step 8
14. Perform a third wash, decant liquid
15. Swirl pellet in remaining liquid and pour into an ice cold 50 mL tube
16. Rinse the centrifuge bottle with some cold 10% glycerol and add this to the 50 mL tube
17. Add a sterile, cold 10% glycerol solution to a final volume of 40 mL
18. Spin the tube for 10 min at 4200 rmp/ 2°C
19. Pour off the supernatant; estimate the final volume of the pellet and add an equal amount of cold 10 % glycerol (typically 1.3 mL glycerol). Mix gently.
20. Aliquot 50 µL into each chilled microcentrifuge tube and immediately snap freeze in a dry ice/95% ethanol bath
21. Store aliquots at -80°C

Transformation

Preparation

1. Prepare a round-bottom tube with 1 mL SOC for each plasmid
2. Thaw DNA and electrocompetent cells on ice and chill 2 mm cuvettes on ice
   NOTE: *Electrocompetent cells must stay cold!*
3. Set the electroporation machine to the following settings: 400 ohms, 25 µFD, 2.5 kV (volts have to be set manually every time)
4. Label 3 LB/Amp plates for each plasmid (used to grow transformants under selection at 3 different concentrations).

Electroporation

1. Add 1-2 µL of DNA to cold electrocompetent cells
2. Mix the DNA/cell solution by pipetting once, then transfer the solution to chilled cuvette
3. Have a pipette to transfer 200 µL of SOC to the cuvette ready (use P200 – easier fit)
4. Dry the outside of the cuvette with a kimwipe and place it into electroporator
5. Hold down the pulse button until a beeping sound is heard
6. Immediately add SOC (prepped in step 3) to the sample in the cuvette
7. Transfer all contents from the cuvette to the round bottom tube
8. Check the time constant; should be between 8.5-9.5

Culturing Transformants

1. Grow the SOC culture in a shaking incubator at 37°C for 45 minutes
2. Plate 10, 50, and 200 µL of sample culture onto three prelabeled LB/Amp plates
   NOTE*: It is impossible to spread 10 or 50 µL thus bring the volume up to 200 µL using LB/Amp broth; e.g. drop 150 µL LB/Amp broth on the plate and add 50 µL sample into the drop, then spread*.
3. Grow plates at 37°C overnight
4. To confirm the transformation was successful, proceed to DNA miniprep, restriction digest, and gel electrophoresis procedures detailed below

- While most plasmids are ampicillin resistant, occasionally we receive plasmids that are kanamycin resistant. Those transformed bacteria need to be spread and grown on LB/Kanamycin broth and plates.
- Most plasmids are transformed into XL10 Gold cells. However, some plasmids require the DB3.1 host strain. DB3.1 cells are used for propagating plasmids containing the ccdB gene such as in the GATEWAY System.

DNA Miniprep

Below is the protocol adapted from an Invitrogen Quick Plasmid Miniprep kit.

1. Use a toothpick to inoculate a single colony into a round-bottom tube with 2 mL LB/Amp. Grow in shaking incubator at 37°C overnight
2. Centrifuge the culture and decant to completely remove the medium
3. Add 250 µL resuspension buffer to the cell pellet and resuspend until homogeneous
4. Add 250 µL lysis buffer; mix gently by inverting 5 times (do not vortex!!). Incubate at room temperature for 5 min
5. Add 350 µL precipitation buffer; mix immediately by inverting the tube until homogeneous (do not vortex!!); centrifuge the lysate at >12,000 x g for 10 min using the microcentrifuge
6. Load the supernatant from step 5 into the spin column in a 2 mL wash tube; centrifuge at >12,000 x g for 1 min and discard the flow through; place column back into wash tube
7. Add 700 µL wash buffer (W9) to the column and incubate for 1 min at room temperature; centrifuge the column >12,000 x g for 1 min; discard flow through and place column back into wash tube. Centrifuge column one more time >12,000xg for 1 min to remove all liquid; discard wash tube with flow through
8. Place spin column into a clean recovery tube; add 75 µL preheated TE buffer to center ofcolumn for elution; incubate column for 1 min at room temperature
9. Centrifuge column with recovery tube at >12,000 x g for 2 min; the recovery tube contains purified plasmid DNA; discard the column. Use DNA immediately or store at 4°C short-term (within the week) and -20°C long-term

Restriction Digest

1. Determine what enzyme(s) to use by checking the plasmid map and estimate expected band sizes. Determine the appropriate 10x buffer for the enzyme or enzyme combination. The color of the enzyme tube's cap coordinates with the color of the buffer cap. For a double digest, find the appropriate buffer using the following web site: https://www.neb.com/tools-and-resources/interactive-tools/double-digest-finder.
2. Add ingredients to the tube in the order shown in table 6.

Single Digest	Double Digest
2 µL Buffer	2 µL Buffer
12 µL H_2O	11.5 µL H_2O
5 µL DNA	5 µL DNA
1 µL Enzyme	0.75 µL Enzyme 1
	0.75 µL Enzyme 2

Table 6. Single and double restriction digest material amounts

3. Tap slightly to mix and briefly spin if splattered on wall.
4. Incubate at 37°C for 1 – 4 hours
5. Run a gel with the digest immediately or store the digest at -20°C until the gel can be run

Gel Electrophoresis

Materials

- DNA digests
- Gel Comb
- Glass Slides
- 1X TAE Buffer
- Invitrogen UltraPure Agarose
- Invitrogen SYBR Safe DNA Gel Stain
- Thermo Scientific O'GeneRuler 1kb Plus DNA Ladder
- Thermo Scientific Orange DNA Loading Dye 1% Agarose Gel (50 mL)

NOTE: *This recipe makes 4 micro gels, with 8 wells each. Typically, no more than one or two gels are required, adjust recipe accordingly.*

1. Add 0.5 g agarose and 50 mL 1X TAE Buffer into a 125 mL flask
2. Microwave the mixture until the agarose completely dissolves (usually a little over a minute, let media bubble for just a few seconds)
3. Let liquid gel cool down in a 60°C water bath, or on the bench when slightly swirling from time to time
4. When liquid is cool enough to touch add 5 µL of DNA gel stain and slightly swirl (no bubbles!)
5. Use a 10 mL pipette and add 12 mL of stained liquid agarose to 1-4 glass slides that are aligned with a comb, as shown in *Figure 12*.
6. Let the gel solidify
7. Add a few drops of 1X TAE buffer onto the gel around the comb teeth to facilitate removing the comb
8. Store unused gels wrapped in Saran wrap in fridge for up to 2 weeks

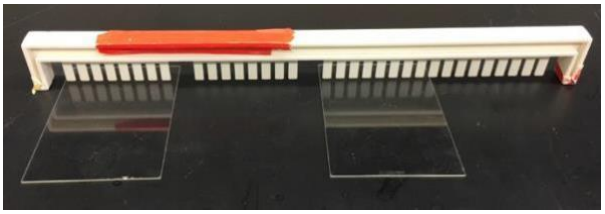


Figure 13. Glass slides with an aligned comb for gel preparation

Running Gel

1. Add 100 mL 1X TAE Buffer and slide the gel with glass into the electrophoresis machine
   NOTE: *Set up the gel to migrate to the positive end as shown in picture.*
2. Add 2.5 µL 1 kb ladder into the first lane
3. In a microcentrifuge tube, mix 2 µL loading dye and 10 µL DNA digest. For uncut DNA, add 1 – 2 µL
4. Add 10 µL of loading dye/DNA mixture into each lane
5. Run the gel at 80 volts for about 30 min (gel is complete when the orange stripes of the loading dye reach the bottom 1/3 of the gel)

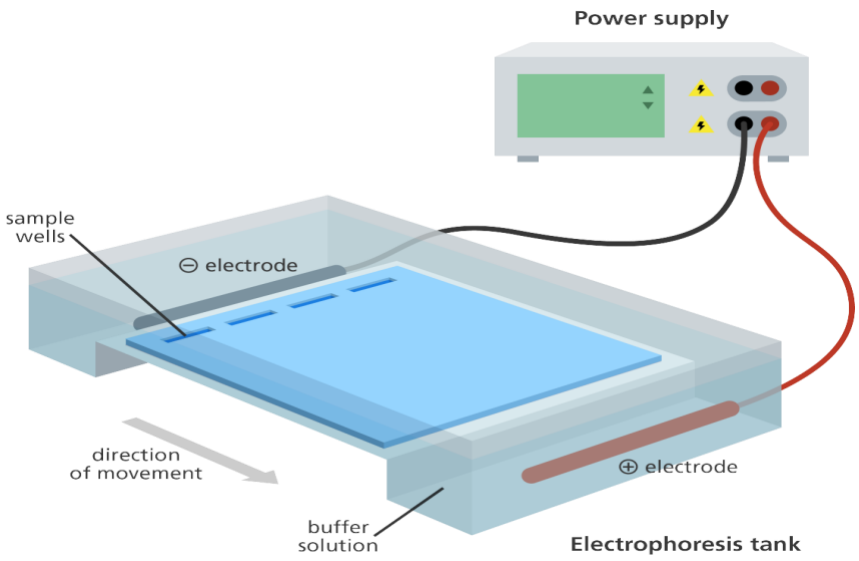


Figure 14. Gel electrophoresis setup

6. Carefully take the gel out of the buffer and observe it under UV light
7. Take a photo with a phone of illuminated gel for documentation

eLab Inventory

We use this inventory system to manage all types of materials held at the stock center: Dicty strains, plasmids, bacterial food sources, and additional material. This inventory system covers Freezer A, Freezer B, Fridge, -20 Freezer, Mini -20 Freezer and Tanks A, B and C.

A majority of questions can be answered by the 'Help' section of eLABInventory.

https://us.elabinventory.com/doc/

NOTE: *Currently, eLabs is not connected to the new dictyBase, but the two will be connected in the next edition of the database via the eLabs API.*

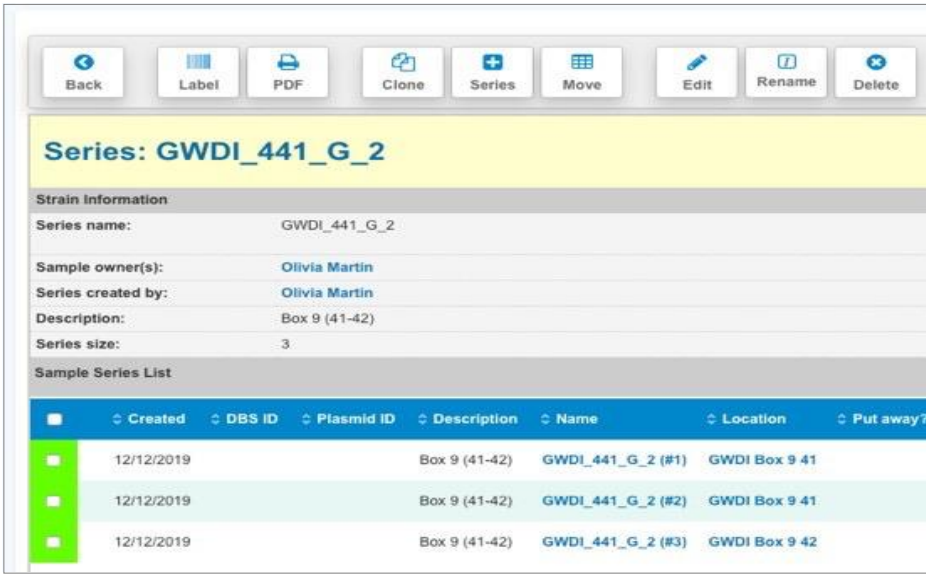


Figure 15. Example strain profile in eLab

Creating Material Profile

1. From Dashboard, select Sample List
2. Select Add
3. Select appropriate sample type (ie. Strain)
    a. Create a series if more than one sample will be stored / exists.
        i. Enter number of samples in series
        ii. Select 'Use incremental name'
4. Fill in the following information for profile:
    a. Name, tube color, date stored, stored as, and stored by
    b. The other boxes: description, tested, test, result, and put away may be filled out later in the storage process

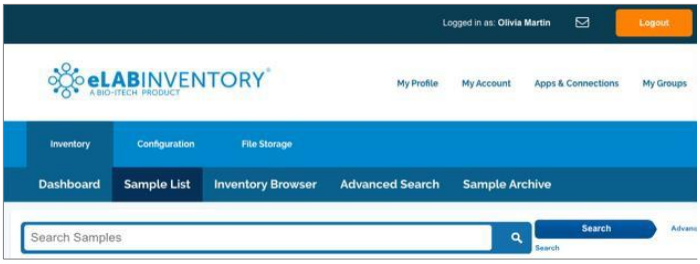


Figure 16. Inventory part of eLAB inventory dashboard

NOTE: Each sample type profile has different fields to add information to.

Labels

We use the Cryogenic Barcode Labels – 1″ x 0.75″ (SKU: JTTA-4) from GA International.

Printing a Label

1. Login to eLabInventory.com and select Inventory tab
2. Go to material's profile via the Sample List subtab, the Inventory Browser subtab or by searching sample name
3. If the sample is part of a series, make sure individual names have the following format [sample name] (#1)
4. Click on Label tab to print label

General ZPL Template

```
^XA
^MMT
^PW203
^LL0153
^LS0
^FT1,200^BQN,2,4
^FH\^FDLA,{{{barcode}}}^FS
^FT1,90^FB203,4^A0N,20,19^FH\^FD{{{name}}}^FS
^FT90,120^A0N,19,19^FH\^FD{{{Date Stored}}}^FS
^FT90,145^A0N,22,22^FH\^FD{{{Stored as}}}^FS
^PQ1,0,1,Y^XZ
```

Label Size: 2.15 x 1.25 inches

NOTE: Labels can only be printed on the Windows computer, which connects to the Zebra Printer.

Create/Adjust Storage Units

1. Select Configuration
2. Select Storage Units
3. Select Add Storage Unit to add new storage units

GITHUB

Add the appropriate label for the payment method

PO Pay
Credit Card Pay
Wire Transfer Pay
Chart String Payment

DSC Data Curation

DSC strains and plasmids are added by submitting annotations in table form. Tab-delimited tables are uploaded, and data will be updated periodically in our emerging database. Tables are uploaded in GitHub.

At a later time all strains and plasmids will be edited via HTML5 inline editing. The authorized DSC user logs in via third party account and the DSC relevant fields will be accessible for editing.

In the strain section on the gene page, each current strain is editable and a + sign allows to add a new strain. Strain annotation fields that appear for annotation, such as strain descriptor and genotype (manual), and mutagenesis method or strain characteristics (form ontologies). Wild type strains that do not refer to a gene, are added via the strain catalog page.

Plasmids are equally added via this route. Plasmid annotations that are specific for a gene, are added on the gene page and thus automatically linked to the gene. Plasmids that do not refer to any gene, are added via the plasmid catalog.